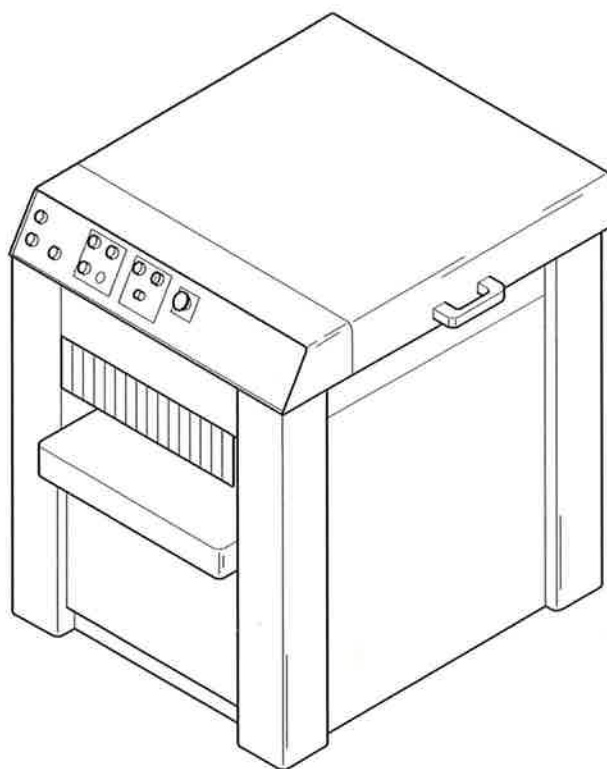


RS53S - RS63S



USE AND MAINTENANCE INSTRUCTIONS



Edition 07/2000

Code 62900024

EC CONFORMITY DECLARATION

SUERI ALFREDO S.p.A.
Via Carpi Ravarino, 115
41010 Limidi di Soliera (MO) - Italy

Declares that the **THICKNESS PLANER**

Model	RS 63 S
Serial number	AB1631 AB1642
Manufacturing year	2001

complies with following provisions:

DIRECTIVE 98/37/EEC:

Basic requirements for the safety and health protection applied to the machines.

DIRECTIVE 73/23/EEC and later modifications:

Safety guarantees of the electric material to be used within certain voltage limits.

DIRECTIVE 89/336/EEC and later modifications:

Electro-magnetic compatibility

Applied harmonized norms:

UNI EN 292-1	EN 953	UNI EN ISO 11202
UNI EN 292-2	UNI EN 954-1	UNI EN ISO 11204
UNI EN 294	UNI EN 981	CEI EN 60204-1
UNI EN 349	UNI EN 1037	CEI EN 50081-2
UNI EN 418	UNI EN 1088	CEI EN 50082-2
UNI EN 860	UNI EN ISO 3746	

Approving institute:

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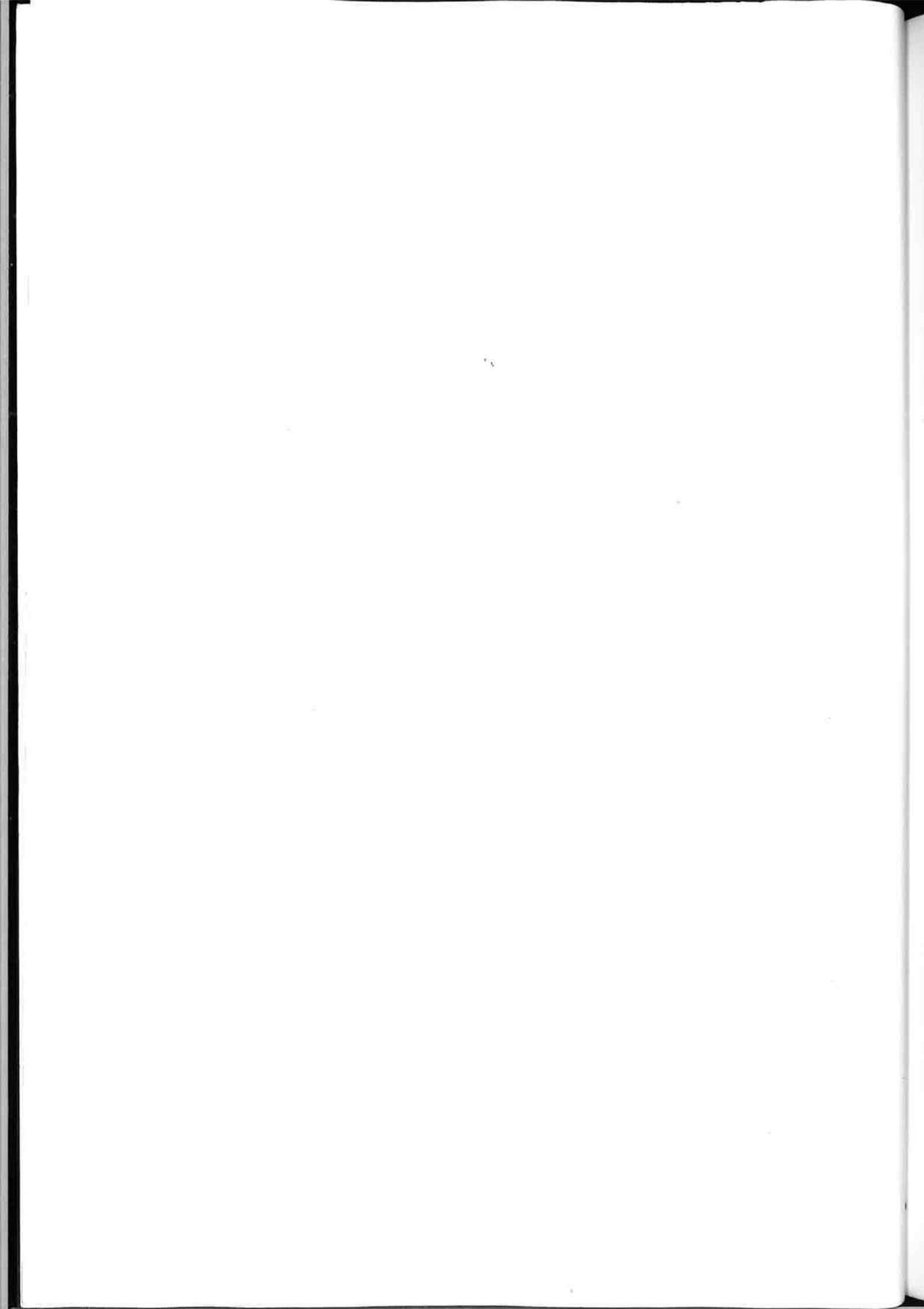
CE sample examination Conformity certificate number: 9 8 1 2 1 7

Limidi di Soliera,
on

The Production manager
Mauro Menni

25/07/01





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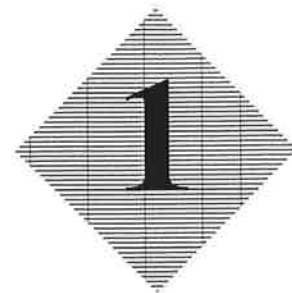
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1 INTRODUCTION

1.1 MACHINE DESCRIPTION

Machine designed for flattening the surface of wood and similar materials by a horizontal rotating tool placed on the table used for the piece positioning and support. The material thickness depends on the removal of the chips from the piece upper surface.

A piece automatic feeding device is enclosed on the machine, and the piece is held and controlled mechanically by it during the machining; an anti-rejection device is also enclosed, which reduces the possibility of rejection and ejection of the piece.

The machine can have one or more of the following configurations:

- a)** automatic table positioning by a ratiomotor with value reading on an electronic display;
- b)** continuous variation of the feeding speed with electronic device;
- c)** the piece feeding device includes two rollers, one before and one after the planer shaft; a third roller at the exit can be installed upon request;
- d)** the table is usually smooth, but it can also be equipped with rollers aiding the piece feeding.

1.2 MACHINE IDENTIFICATION

1.2.1 Machine identification

The machine is identified by the writing indicated in the plate (Fig. 1) on the basement, as shown in Fig. 2:

Constructore/Hersteller/Manufacture/Constructeur	
Type	
Nr./Serienummer/Serial No. Nº, d'identification	
Anno di costruzione/Anzahl/Year of construction/Année de fabrication	
La macchina è con Max. Werkzeug o Filo, tool o Outillage	Diametro/Ø/mm Outer diam. mm
Velocità Drehzahl Speed Vitesse	giri/min U/min rpm tr/mn
Tensione/Spinning Voltage/Tension	V Hz 3-
Intensità di corrente/Amperom Current rating/Intensité	A
Schema elettrico Nr./Schaltplan Nr. Electrical drawing No./Schéma électrique No.	
	

Fig. 1: Machine identification plate

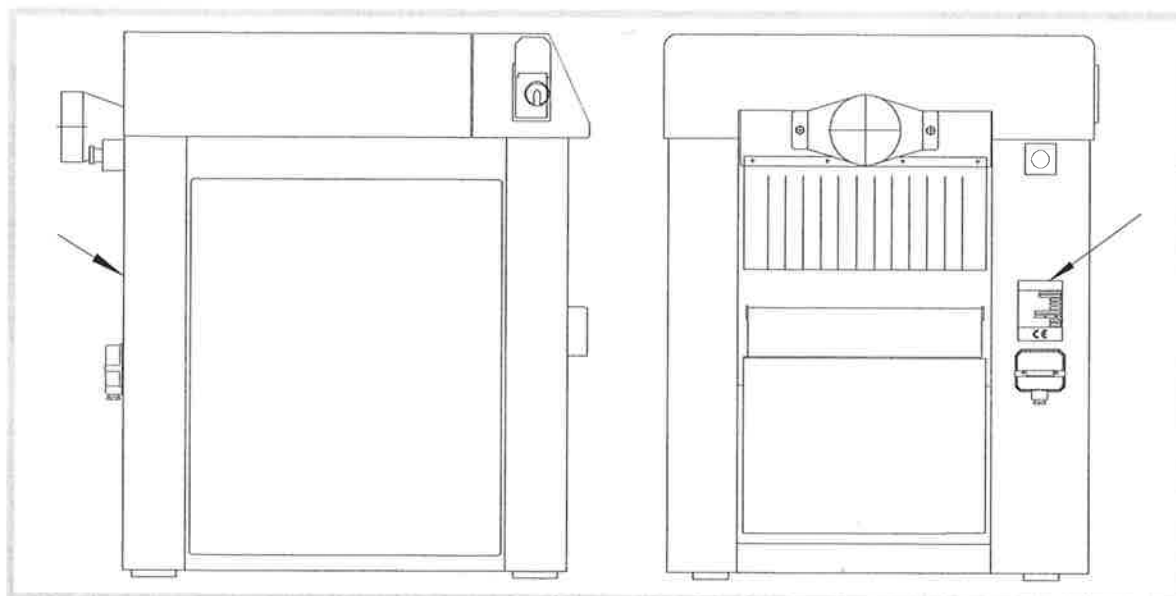


Fig. 2: Machine identification plate

1.2.2 Correspondence

When communicating with the Dealer or the Technical Service Center or the Manufacturer, for any reason concerning the machine, always refer to the following data:

- Machine model
- Machine serial number
- Manufacturing year

1.2.3 Service recommendations

When preparing this instruction handbook, all the operations involved in the routine maintenance have been considered.

We recommend you not to avoid any repair or intervention not indicated in this handbook.

All the operations needing to remove some parts must be carried out by qualified and/or authorized technical staff.

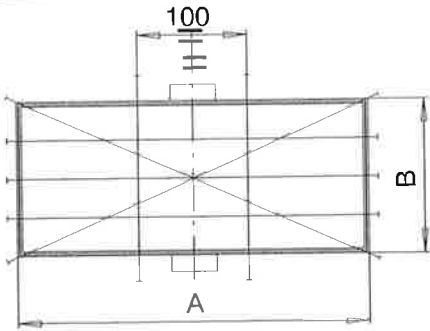
Only skilled and trained staff is authorized to use the machine and to carry out the maintenance operations after having read this handbook.

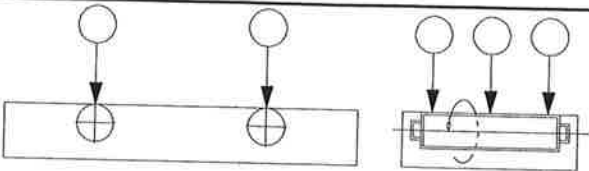
As for technical problems for the Service, give the following information as well:

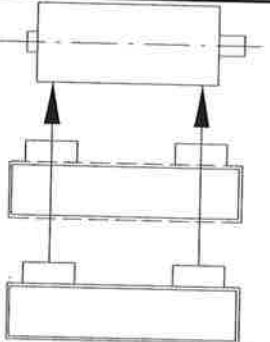
- Connection voltage
- Frequency
- Purchase date
- Detailed information on the possible failure found
- Detailed information on the special machining to be made
- Use period / operating hours

1.2.4 Final test report

The factors for the final test of the machine geometry are referred to the norm ISO 7568, and particularly:

Machining table flatness		
	Tolerance	
Longitudinal (A)	0.30 mm	
Diagonal (A)	0.30 mm	
Transversal (B)	0.15 mm	

Radial swinging		
	Tolerance	
	0.15 mm	
Parallelism of the rollers on the table:		
	Tolerance	
	0.15 mm	

Parallelism between table and planer shaft		
	Tolerance	
	0.15 mm	

1.2.4.1 Positioning of the feeding rollers and pressers to the cutting circumference

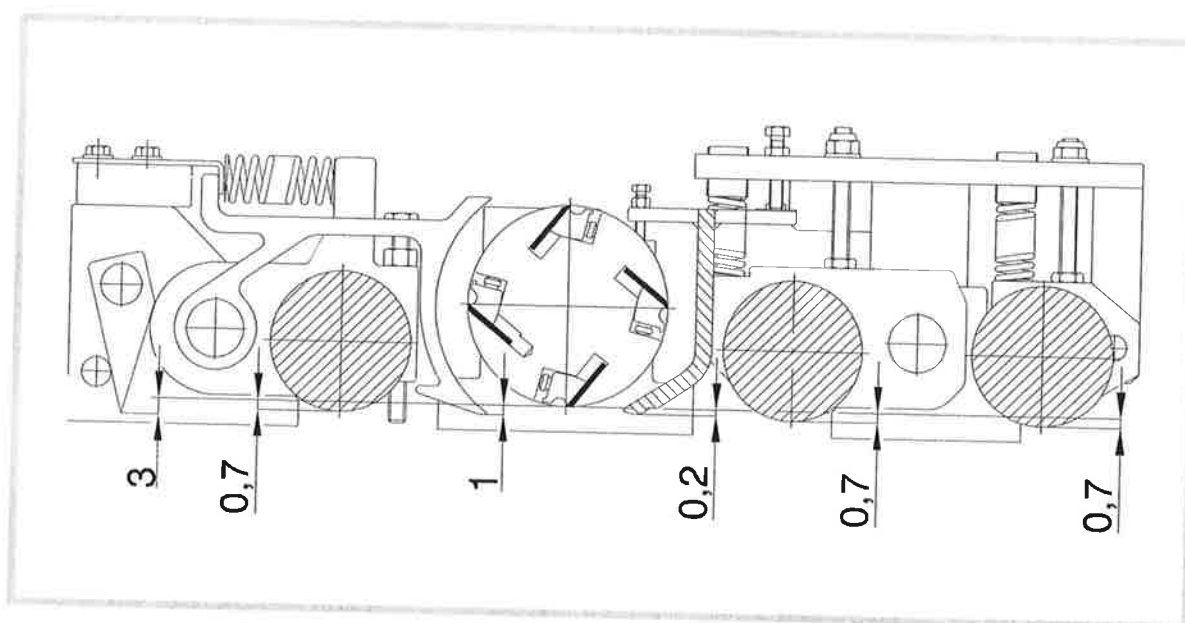


Fig. 3: Position of the steel rollers

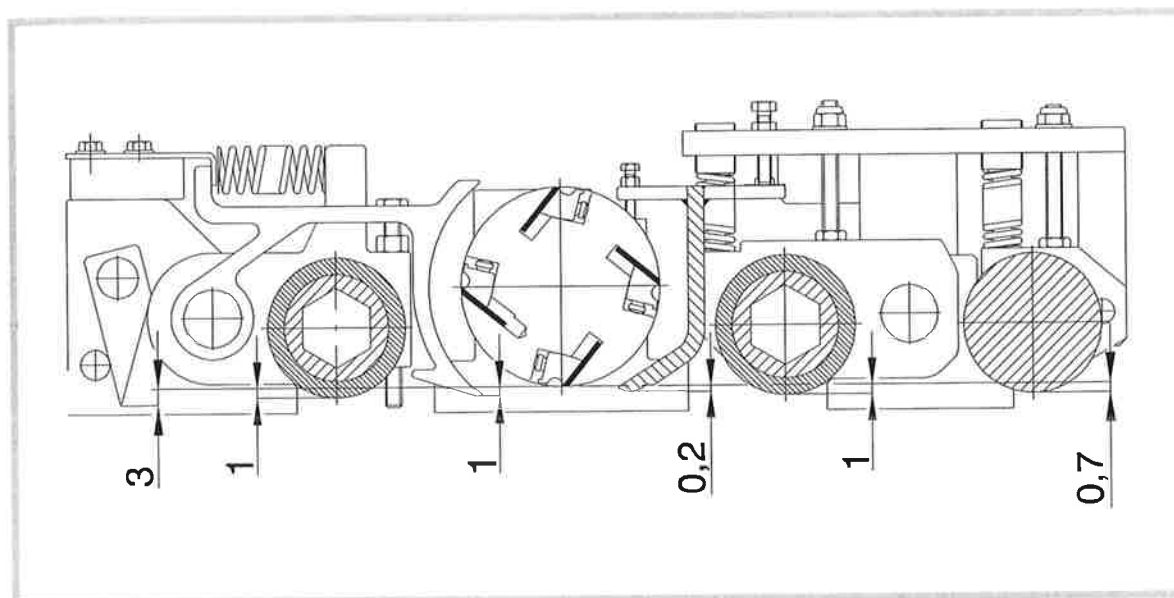


Fig. 4: Position of the rubber-coated rollers

Seller's/dealer's stamp

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2 WORKING SAFETY

2.1 GENERAL SAFETY RULES

The machine operator and the person in charge must read this instruction handbook carefully and completely before executing any operation on the machine.

Keep this handbook close to the machine.

The manufacturer is continuously improving the products, so that some machine components may possibly change.

All the technical information in this handbook are exclusive property of the manufacturer and are strictly confidential.

Thus, it is forbidden to reproduce and disclose them, even partially, without a written authorization by the manufacturer.

It is also forbidden to use this handbook for purposes other than those strictly connected with the installation, use and maintenance.

For detailing shortly and clearly the operation, adjustments and other, some figures may show the machine without safety guards. **DO NOT USE** the system under these conditions, and in case of maintenance or repair interventions follow the indications in the chapter: **MAINTENANCE**.

2.1.1 Operator training

It is extremely important that all the operators are suitably trained in the use, adjustment and utilization of the machine.

Above all:

- a) the principles for adjusting and using the machine, including a correct use and adjustment of the devices for guiding and holding the piece, guards and tools.
- b) The piece safe handling while machining, following the recommendations for using templates and similar devices:
 - sloped planes and bevels can be machined using templates to support the piece and produce the wished angles;

- arrange sturdy edges to prevent side movements of the piece;
 - equip the template base with edges on both sides of the machine table, to effectively hold the piece during the feeding;
 - use roller devices or extension tables when long pieces are machined.
- c) The safe storage of pieces before and after the machining.
 - d) The correct adjustment of the safety devices.
 - e) Start the machining only when the tools have reached the operating speed.
 - f) The use of personal protections for ears, eyes and lower limbs.

This machine has been designed and manufactured to give the user suitable safety but, like all machines provided with cutting systems, it may cause dangerous situations due to incorrect use of the machine, lack of experience or tampering with the safety devices of the machine; therefore the operators should take these remarks into account and carefully follow the instructions given in this handbook about safety.

2.1.2 Symbols

To point out the handbook sections concerning personal safety, the following symbols have been used in the USE AND MAINTENANCE operations of the machine:



ATTENTION

Information or procedure that, if not carefully followed, could cause death or serious personal injuries.



CAUTION

Information or procedure that, if not carefully followed, could cause light personal injuries or machine damages.



ELECTRIC DANGER

Information or procedure that, if not carefully followed, could cause death or serious personal injuries.



WARNING

Information or procedure that aids the operator in the best use of the machine or equipment, to lengthen their life, avoid damages or programming losses, optimize the work in compliance with safety norms.



NOTE

Important information or procedure.



It is recommended to use the personal protection devices for the eyes.



It is recommended to use the personal protection devices for the ears.



It is recommended to use the personal protection devices for the hands.



It is recommended to use the personal protection devices for the lower limbs against the squeezing due to falling pieces during the operation.

2.2 MACHINE CORRECT AND IMPROPER USE

The machine can carry out flattening operations on deadwood, veneered wood, plastic materials with features similar to wood, with a max. removal of 8 mm.

Any other use or machining on a material other than the indicated ones (for example metals or materials with similar features) is forbidden, as it can be risky for the operator.

The manufacturer will not be responsible for damages or injuries to people and/or things caused by a use non-complying with the norms. The risk will exclusively be at the user's charge.

This machine cannot operate in explosive environments.

2.2.1 Number and position of the working places

For a correct and safe use of the machine/one operator is expected at the entry of the material to be machined, and possibly one operator at the exit to avoid piece falling and smashing.

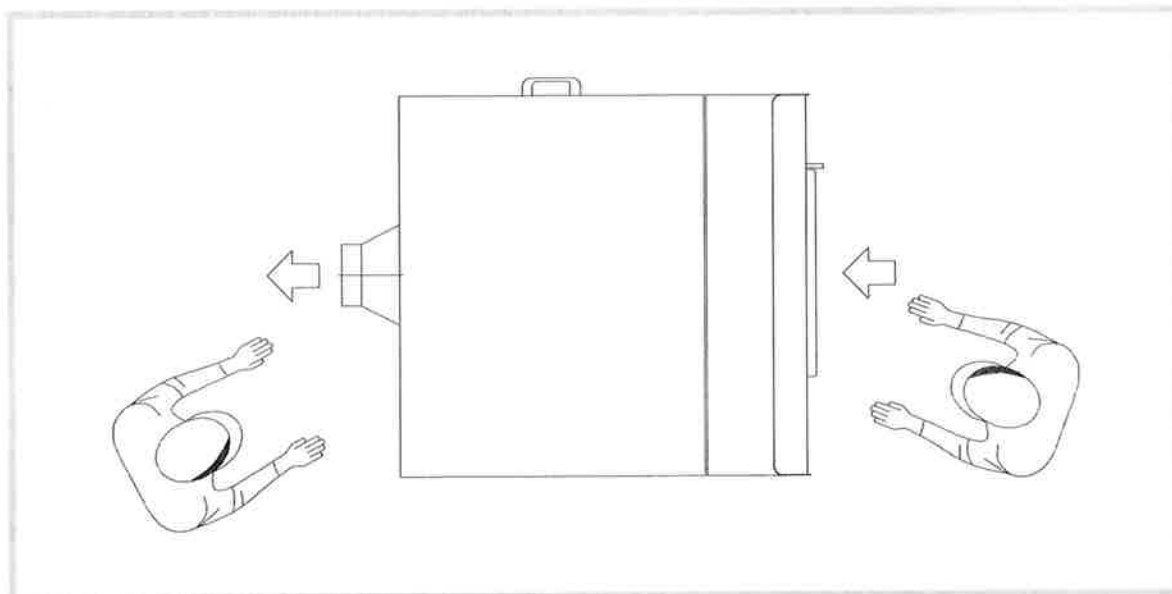


Fig. 5: Working places

2.3 INDIVIDUAL RISK FACTORS

Some clothes or accessories worn by the operator can cause accidents. It is advisable to avoid wearing dangerous objects, such as: rings, watches, bracelets, large-sleeved overalls, not perfectly fastened belts, ties and, generally, anything sticking out of the person that may be grasped by a moving part of the machine during its operation: people with long hair should keep it gathered on the back of the neck, and possibly wear a cap.

People not belonging to the work place shall absolutely stay far off the operating machine.

2.4 ENVIRONMENTAL RISK FACTORS

The place of work must be properly illuminated.

The operations performed by the machine produce dust and, in general, the machine is in a place where other machines spread dust in the environment; to avoid dangerous dust concentrations it is necessary:

- to make a correct connection with the suction system;
- to periodically carry out the maintenance on the suction system filters;
- further, as the dust deposited on the floor generates danger conditions, it is necessary to carefully clean the place of work close to the machine periodically.

2.5 RISK FACTORS CONNECTED WITH THE MACHINE USE

- When the machine is connected to the mains, check the correspondence between the electrical features of the machine and those of the mains; it is also advisable to check the correct grounding of the machine; these steps must be recalled to the specialized technician carrying out the electrical connection.
- Before carrying out any maintenance on the machine, switch it off and disconnect the power sources, set the main switch on zero and lock it.
- Machining pieces that are too small or too big for the machine capacity may cause a dangerous situation for the operator or the people near the machine; always compare the sizes of the pieces being machined with the max. and min. dimensions that the machine can work.
- Do not remove or change the protection devices to change the operating capacity of the machine.
- Be very careful when choosing and maintaining the tools.
- Never use cracked, strained or excessively worn tools, as their utilization impairs the machining quality and can involve risks for the people nearby the machine, as well as further increase the machine noise.
- Always wear gloves when handling the tools; refer to the data in the handbook for the knife locking and projecting, as well as for the best adjustment of the knife locking screws.
- A correct maintenance will ensure higher working safety as well as a constant quality level of the machining in time.

- The protection devices must be regularly checked for their operation, for example before every working shift.

2.6 RESIDUAL RISKS

Even with the guards installed and operating correctly, the machine can originate the following residual risks:

- breaking of a tool while machining;
- breaking of the piece being machined;
- electric shock;
- noise exposure, above all if the indicated personal protection devices are not used;
- emission of dust, above all if the suction system is not correctly connected and maintained.

2.7 PROTECTION AND SAFETY DEVICES

The machine is equipped with the following protection and safety devices:

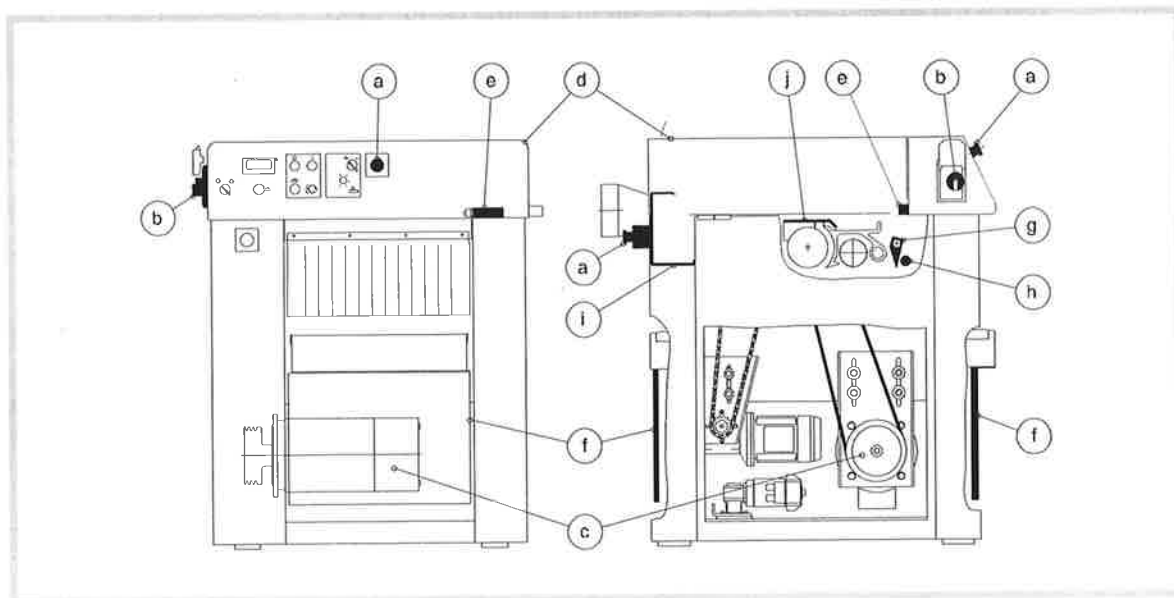


Fig. 6: Protection and safety devices

- a) two emergency stop push buttons, one on the piece entry side and one of the exit side of the machine;
- b) lockable main switch;
- c) self-braking motor with intervention lower than 10 s;
- d) interlocked upper cover, it stops the spindle rotation when opened;

- e) cover interlock micro-switch;
- f) front and rear guards on the table;
- g) piece anti-ejection device with anti-return arms, automatically operating with any wood thickness, that avoid the piece being machined from being thrown towards the operator, thus ensuring an efficient active protection;
- h) passage limiting bar;
- i) piece anti-falling device;
- j) planer shaft guard.

It is important to check the correct operation of the safety and protection devices as indicated in § 5.1.2.

2.8 ENVIRONMENTAL SAFETY

2.8.1 Noise emission

The operating conditions for measuring the noise meet the enclosure D of ISO 7960: 1995.

The acoustic power levels have been measured according to the measuring method using the developed surface shown in EN ISO 3746: 1995.

The acoustic emission pressure level at the operator place has been measured according to EN ISO 11202: 1996.

Standard spindle	RS53S		RS63S	
	Empty without suction	Operating with suction	Empty without suction	Operating with suction
Emitted acoustic power level dB(A)	89.5	95.9	91.2	96.8
Acoustic pressure level in the working place dB(A) piece ENTRY	78.0	83.7	80.0	84.1
Acoustic pressure level in the working place dB(A) piece EXIT	72.4	79.6	79.3	83.2

Inaccuracy constant of the measurement K = 4 dB.

The reported noise levels are referred to "free field" conditions, in compliance with the test methods expected in the reference norm, UNI EN 860: 1998.

The indicated noise values are emission levels and not necessarily safe levels. Although there is a relation between emission levels and exposure levels, this cannot be reliable for stating whether further precautions are needed or not. The factors influencing the real exposure level of the operators include the features of the working place and other noise sources, such as for example the machine number and other nearby machining. Further, the allowed exposure level can change in different countries. How-

ever, this information will allow the machine operator to better evaluate the danger and the risk.

Some factors reducing the exposure to noise are:

- correct choice of the tool;
- correct choice of the feeding speed;
- maintaining the tools and the machine;
- correct use of the personal protection devices.

2.8.2 Dust emission levels

The dust emission level in the place of work is 0.17 (mg/m³ of air).

2.8.3 Machine removal and demolition

For removing the machine, disconnect it from the power sources (e.g. electric system), clean it carefully and cover the machining tables and the tool-holding shafts with rust preventer. Do not store the machine in wet environment and protect it against weather agents.

2.8.4 Emergencies

In case of flooding of the room where the machine is located, immediately cut the power supply off. Before starting the work again, have the machine checked by a specialized technician.

In case of fire, immediately power the machine off and intervene using suitable fire extinguishers, aiming the jets at the flame bottom. Even if the machine is not apparently damaged, before starting the work again, have the machine checked by a specialized technician.

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3 MACHINE INSTALLATION

3.1 HANDLING - TRANSPORT - STORAGE

Model	RS53S	RS63S
Weight kg	930	1030

Tab. 1: Machine weight

3.1.1 Lifting method

Use lifting means with min. capacity of 1,500 kg.



It is possible to use a lift truck, placing some wooden beams under the basement; in this case, fit some anti-slip material between the forks and the machine basement, taking care to exclude every possible machine overturning.

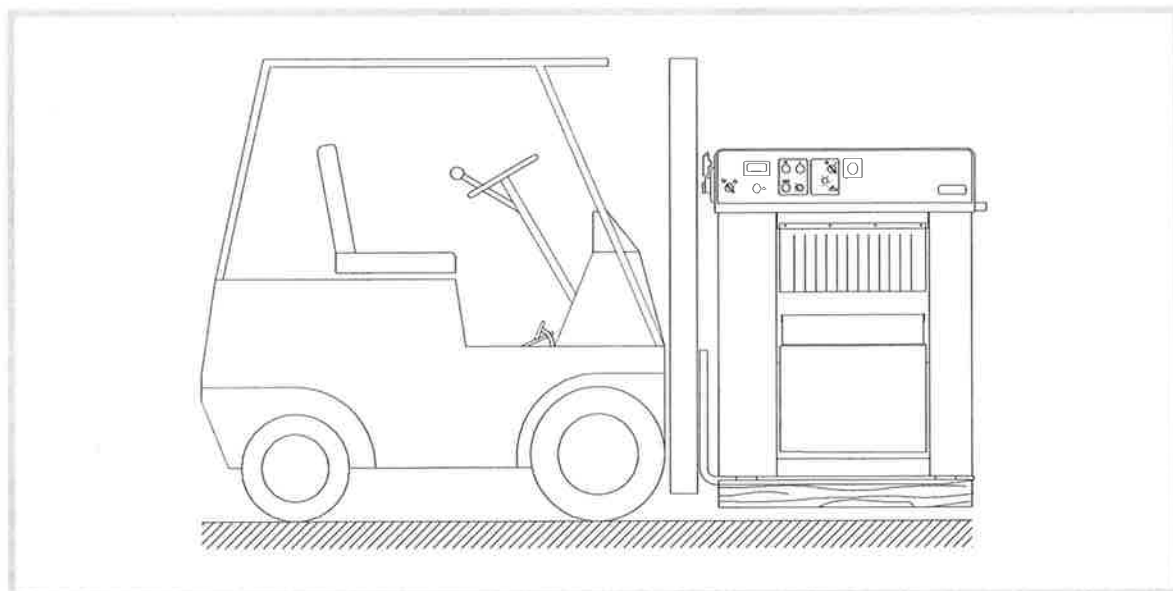


Fig. 7: Machine lifting with lift truck



When using lifting belts or ropes, they must have a suitable capacity and be positioned as indicated in Fig. 8, using the supplied brackets and protecting the machine parts touching the lifting means. While lifting avoid sudden movements or jerks. The max. use load of a lifting rope, belt or chain corresponds to the max. lifting load only if the lifting means are perfectly vertical. In case of inclination, the max. lifting load is reduced according to the indications of Table 2.

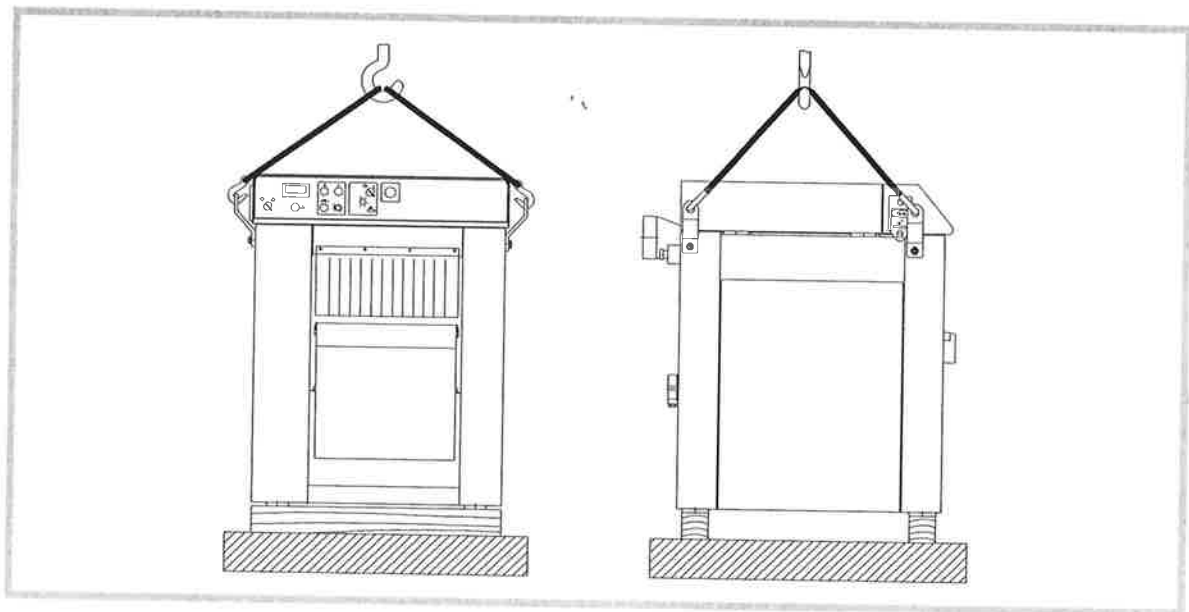


Fig. 8: Machine handling by crane

Angle α	Load reduction
30°	- 15%
45°	- 30%
60°	- 50%

Tab. 2: Lifting load reduction

When the machine moves, do not stay in the manoeuvring area.

3.1.2 Storage

Do not leave the machine under bad weather conditions and grease all oxidizable parts if a long storage is expected, which must be in a dry place and at temperatures between 0°C and 50°C.

3.2 POSITIONING

3.2.1 Positioning and foundations

Place the machine on a reinforced concrete board with a thickness of about 150 mm, properly levelled and sized a little larger than the machine overall dimensions. It is not necessary to fasten the machine to the floor. It is advisable to lay the machine on 4 anti-vibration rubber plates, 5-6 mm thick.

Make the machine stable by levelling it to the floor; in this way, it can be operated efficiently and accurately.

3.2.2 Min. spaces for a safe use

Before placing the machine check if the working area has a sufficient space for enabling its correct utilization.

Leave the area around the machine free from objects that may hinder the operator activity, for example scaffoldings or other fixed structures.

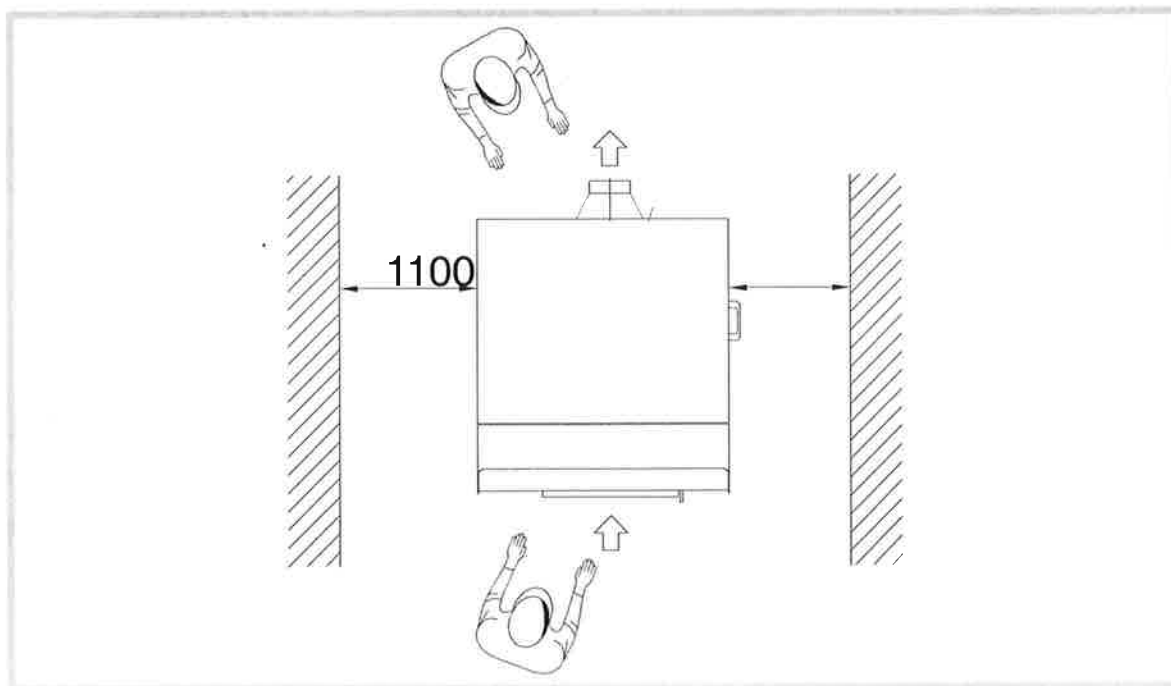


Fig. 9: Machine positioning

3.2.3 Working area lighting

It is recommended to comply with the norms in force as for lighting in the working area with sufficient natural light; in any case the above working premises and places must

be equipped with devices enabling an artificial lighting suitable to protect the safety, health and comfort of the operators.

3.3 MACHINE PREPARATION FOR THE START-UP

3.3.1 Check of the machine possible damaging

At the machine delivery check that there is no damage due to the transport; if there is any damage, immediately inform the forwarder or contact directly the seller and/or the manufacturer.

3.3.2 Machine installation

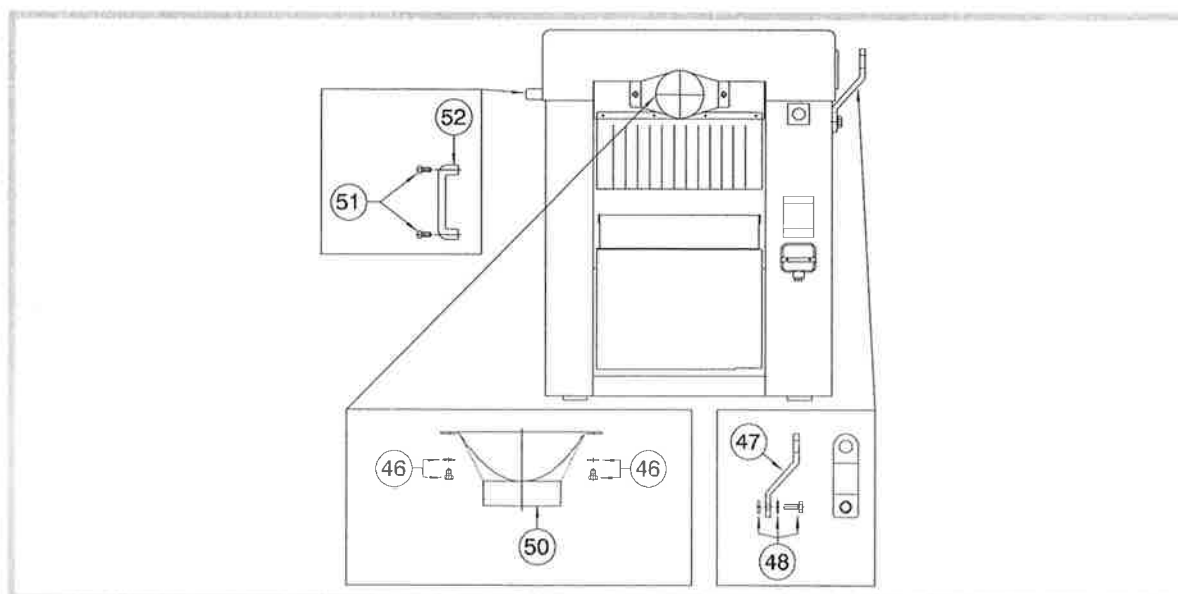


Fig. 10: *Installation of the removed parts*

The machine is delivered with some details removed; they must be installed before starting the machine up:

- install the handle (52) using the screws (51) (for case package);
- install the suction inlet (50) with the supplied screws and washers (46).

With the machine positioned remove the lifting brackets (47), tightening again the screws and the washers (48) in the basement.

3.3.3 Machine cleaning

Before starting to work, degrease carefully the working areas and the guards with a suitable and safe thinner.

3.3.4 Connection with the electric system

All the operations concerning the electrical connection must be made by skilled personnel in compliance with the regulations in force.

Before any electrical connection, make sure the electric line is sized so as to stand the machine power, and check if the mains voltage and frequency correspond to the machine plate data ().



The machine can suit operating voltages included in a tolerance range of $\pm 5\%$, outside of which it is necessary to adjust the supply voltage.

To improve the electrical safety it is necessary to install a lockable switch with fast intervention fuses upstream the machine, AM type.

Refer to Table 3 for choosing the lead and fuse sizes to be used for the machine connection. If the machine distance from the power supply point is long, it is advisable to increase the lead section to reduce the voltage drops along the feeding line.

Absorbed current A	Cable section (3 + G) mm ²	Fuses Type AM
≤ 10	2.5	12 A
$> 10 \div 14$	4.0	16 A
$> 14 \div 18$	6.0	20 A
$> 18 \div 22$	6.0	25 A
$> 22 \div 28$	10.0	32 A
$> 28 \div 36$	10.0	40 A

Tab. 3: Cable section and fuse size

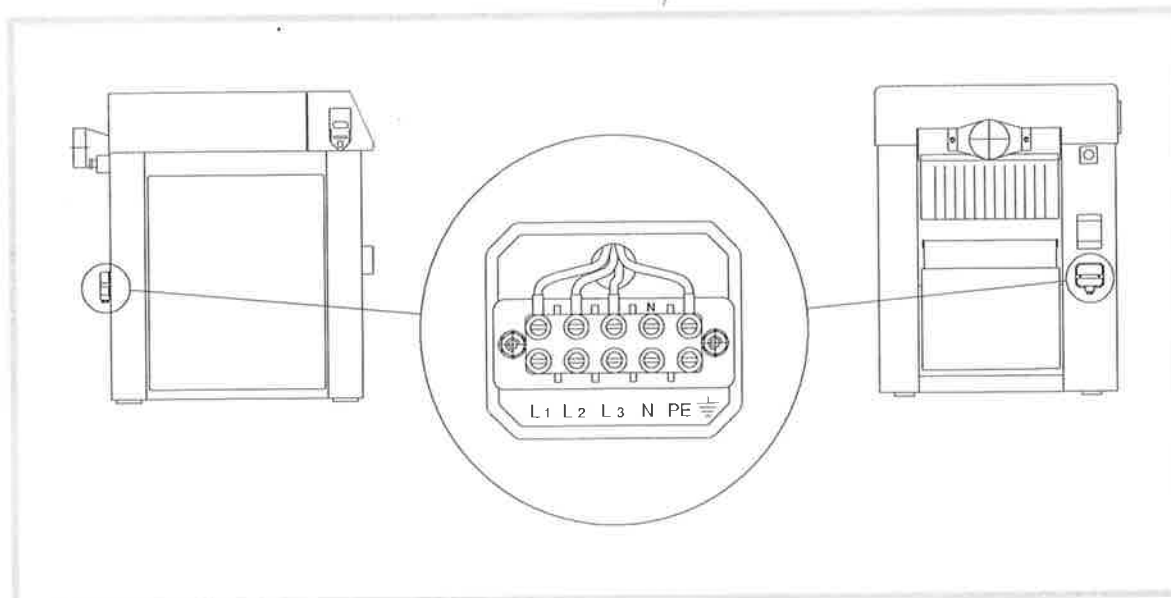


Fig. 11: Terminal board

Connect the three electric cables (phases) to the terminals L1, L2 L3, Figure 11.

Connect the yellow/green cable (ground lead) to the terminal PE and the neutral cable, if present, to the terminal N.

Use N1VV-K type cable for the multipole supply cable, or the type marked H07RN.F for temporary positioning.

Carefully tighten the cable holder and close the terminal board.

Check the rotating direction of the spindle shaft starting the machine as described in § 4.4.1 (the shaft must rotate in the direction opposite to the piece feeding).

If the shaft does not turn in the correct direction, it is necessary:

- to cut the line off;
- to revert two phases of the supply in the terminal board;
- to try again the rotating direction.

3.3.5 Connection with the suction system

It is indispensable to connect the machine to a suction system to eliminate the risks due to dust inhaling and to ensure a safe use and a proper machine operation.

Always operate with the suction system on.

The suction pipes, if in plastic, must be fire retardant.

The machine is supplied with a suction inlet with dia. 160 mm and must be connected with a suction system having the following features:

- capacity: 1800 m³/h (air speed 25 m/s)
- head: 720 Pa

The dust emission in the environment can be reduced by:

- maintaining the tools and the machine;
- maintaining the suction system keeping the filters efficient.

4 MACHINE USE

4.1 CONTROL LIST AND FUNCTIONS

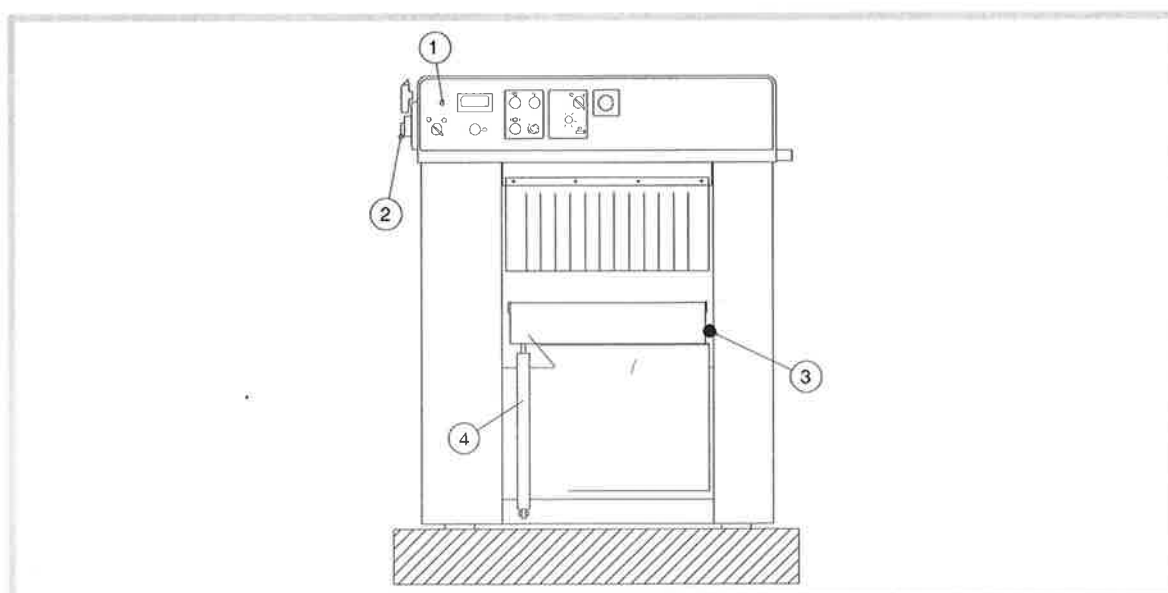


Fig. 12: Machine controls

- 1** - Control panel
- 2** - Lockable main switch
- 3** - Lever for adjusting the rollers on the table (upon request)
- 4** - Table position reading system

4.1.1 Control panel for manual version

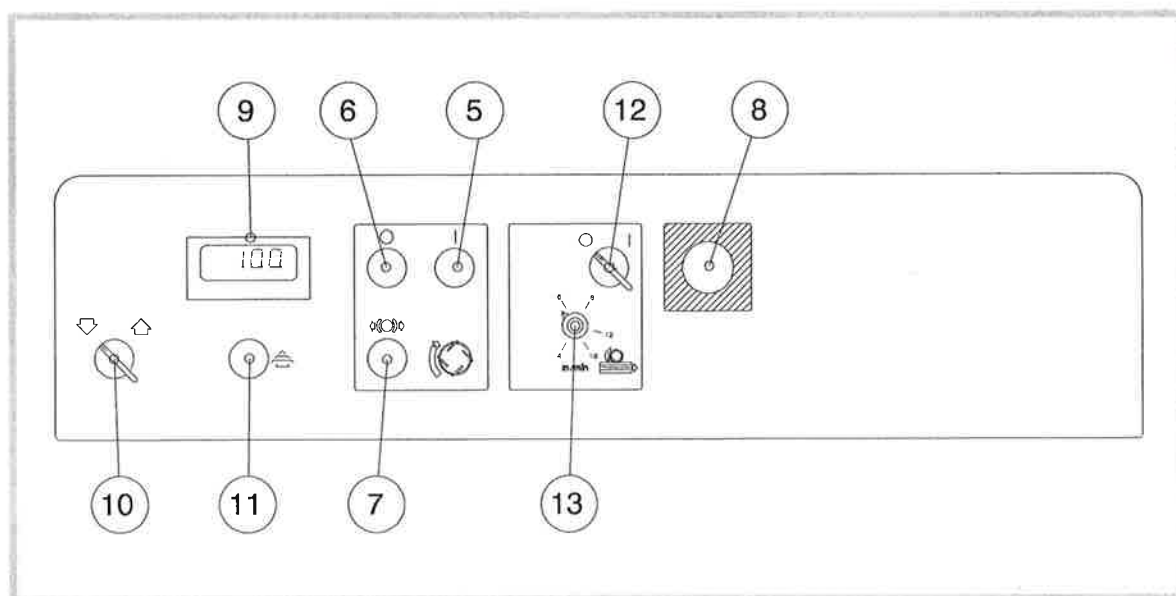


Fig. 13: Control panel with display

- 5 - START push button
- 6 - STOP push button
- 7 - Brake release push button
- 8 - Emergency push button
- 9 - Decimal value display
- 10 - Fast lifting/lowering switch
- 11 - Fine positioning push button
- 12 - Inverter ON/OFF switch
- 13 - Feeding speed variation potentiometer

4.2 PREVENTIVE CHECKS

Before starting the spindle:

- check if the installed tool is suitable for the machining to be made;
- check if the guards for the dangerous areas are correctly positioned and operating;

4.3 INITIAL ADJUSTMENTS AND OUTFIT

4.3.1 Table manual positioning with handwheel

Before positioning the working table check if there are wood residues on the table.

The table adjustment is made operating the switch (10) for the approximate positioning while lifting or lowering and later, for a correct fine positioning while lifting, press the push button (11); an impulse is equal to a movement of about 0.10 mm of the table.

The indication of the machining value is shown on the display (9), expressed in mm and 1/10 mm.

For a correct positioning it is advisable to be always positioned upwards, to cancel the matching clearances between screws and scrolls.

On the piece anti-falling device there are two screws that act as height limit switch; the min. passage thickness is achieved with the table completely lifted.

When a possible wrong manoeuvre locks the machining table against the lower or upper mechanical stops or against the wood, the machine locks the positioning movement. Revert the advancing direction by the switch (10) to restore the standard operating conditions of the machine.

4.3.2 Table automatic positioning with push buttons

Before positioning the working table check if there are wood residues on the table.

The table adjustment is made operating the switch (13) for the approximate positioning while lifting or lowering and later, for a correct fine positioning while lifting, press the push button (14); an impulse is equal to a movement of about 0.10 mm of the table.

The indication of the machining value is shown on the display (12), expressed in mm and 1/10 mm.

On the piece anti-falling device there are two screws that act as height limit switch; the min. passage thickness is achieved with the table completely lifted.

When a possible wrong manoeuvre locks the machining table against the lower or upper mechanical stops or against the wood, the machine locks the positioning movement. Revert the advancing direction by the switch (13) to restore the standard operating conditions of the machine.

4.3.3 Pressure adjustment on the feeding rollers

The spring pressure of the feeding rollers is factory-adjusted during the final testing, in order to get a perfect wood feeding.

Only in special cases intervene and adjust the pressure on each roller, taking care to increase or decrease it equally.

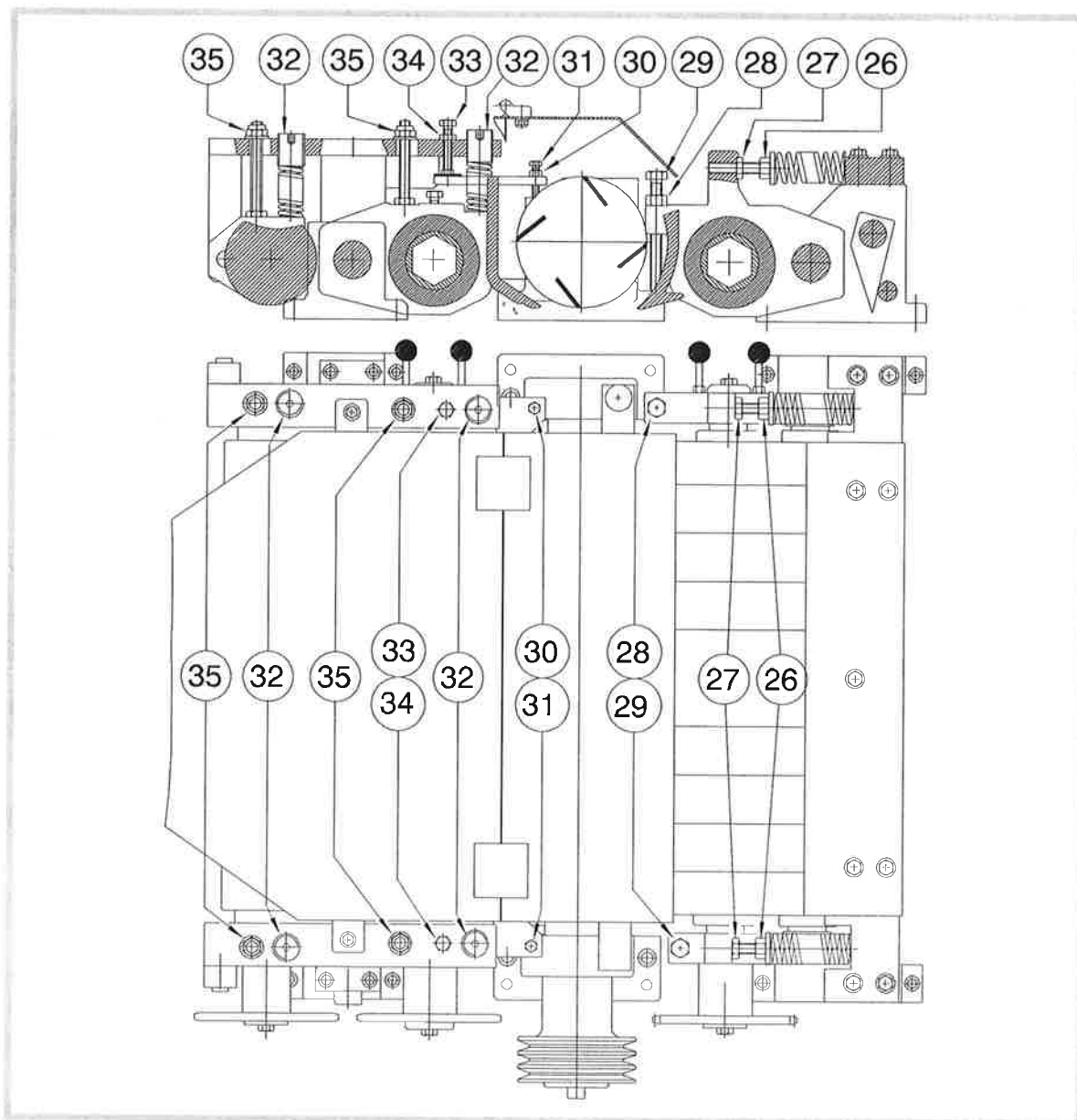


Fig. 14: Pressure adjustment on the feeding rollers

4.3.3.1 1st feeding roller

With the fork wrench, by 17 mm, loosen the nut (27) and tighten the screw (26) to decrease the spring load, loosen the screw (26) to increase the spring load; at the end, tighten the nut (27).

Carry out this step on both roller supports.

4.3.3.2 2nd and 3rd feeding rollers

By the hexagonal wrench, by 6 mm, rotate the pins (32) clockwise to increase the spring pressure, counterclockwise to decrease it.

Carry out this step on both roller supports.

4.3.4 Roller adjustment on the table

To adjust the roller projection from the table, lower the lever (3) to lift the rollers so that they project from the table by 0.2 mm at most.

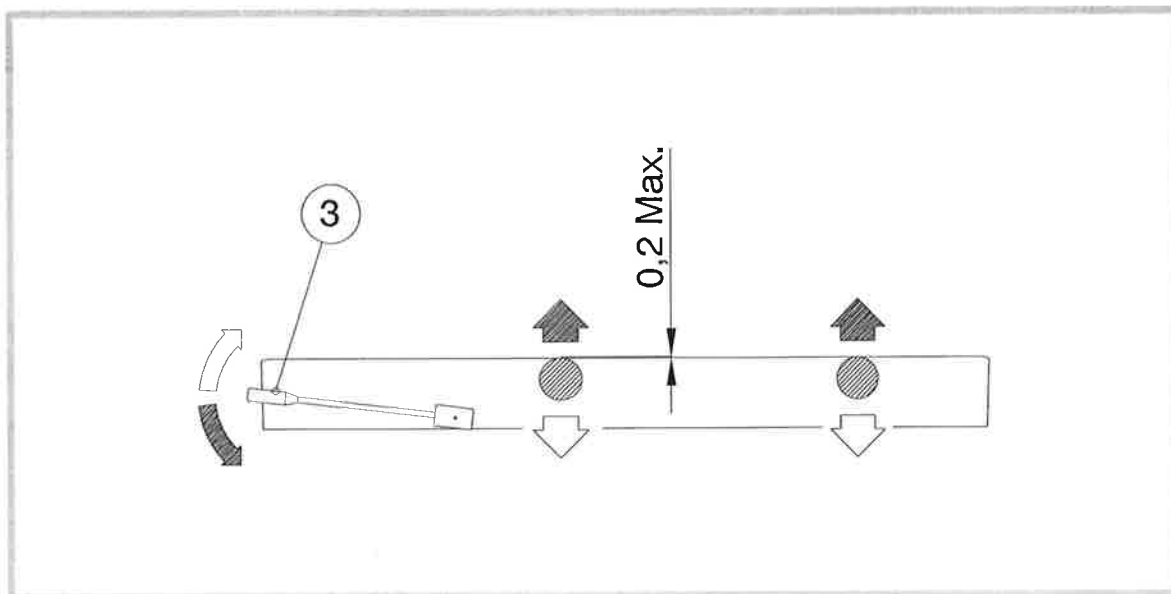


Fig. 15: Roller adjustment on the table

4.4 OPERATING CYCLE

4.4.1 Operating procedures

The procedure below refers to the machining of standard shape pieces, with dimensions compatible with the machine features.

If pieces with a special shape are to be machined, it is necessary to realize some machining equipment as described below in § 4.4.3.

Before proceeding with the table adjustment it is necessary to measure correctly the thickness of the piece/s being machined; consider that:

- a) the max. removal is 8 mm;
- b) when several pieces are machined simultaneously the max. thickness difference must not exceed 2 mm, and only in case the first feeding roller is sectioned;
- c) do not feed tapered pieces if you do not use a suitable machining equipment;
- d) the piece must not have irregularities that may make the machining dangerous, inclusions of materials different from wood (e.g. steel nails), and must be pushed with the hands without danger during the machining;

- e) the anti-ejection arms must be in the machining position;
- f) adjust the table height as indicated in § 4.3.1;
- g) start the spindle motor pressing the push button (5);
- h) select the feeding speed by the potentiometer (13) with the spindle motor operating;
the speed must be chosen considering that increasing the piece width or thickness to be removed the speed must decrease;
- i) feed the piece checking if the machining occurs regularly; if the table has rollers adjust them so as to aid the piece feeding.

4.4.2 Flatting of long pieces

For flatting long pieces, more than what indicated in Tab. 4, it is necessary to use some supports applied at the entry and at the exit of the machining tables as indicated in Fig. 16; this allows to hold the piece perfectly stable on the machine working table.

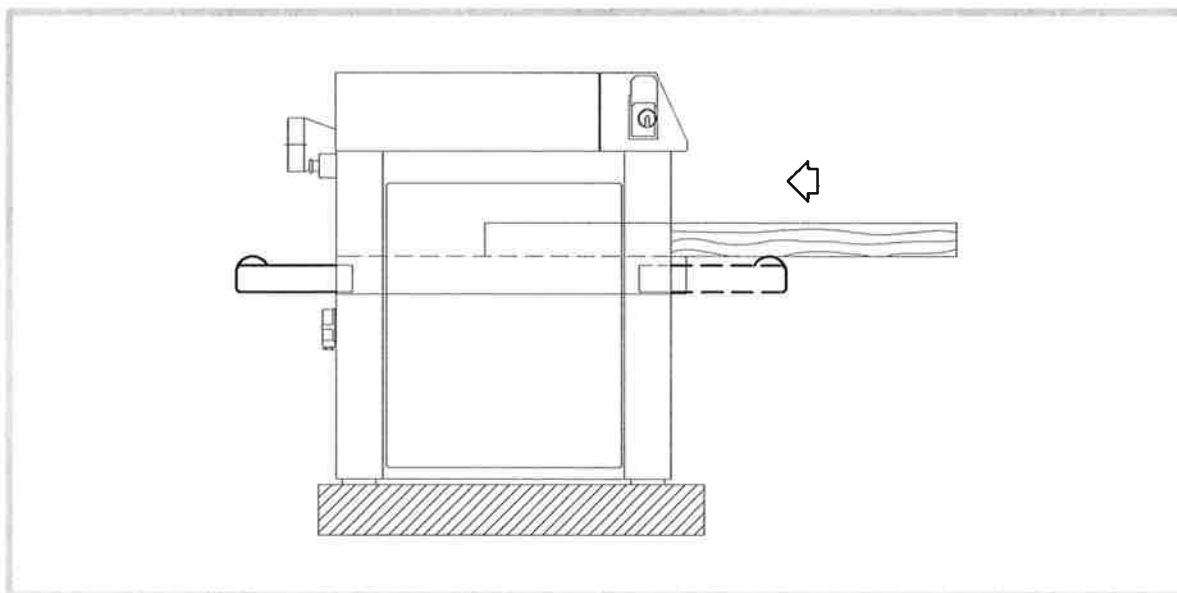


Fig. 16: Machining of long pieces

4.4.3 Flatting of pieces with sloped planes

It is necessary to use jigs with lips in the edges so as to avoid the piece side movement during the feeding (Fig. 17).

In other cases it can be necessary to equip the base of the machine plane to effectively restrict the piece during the machining.

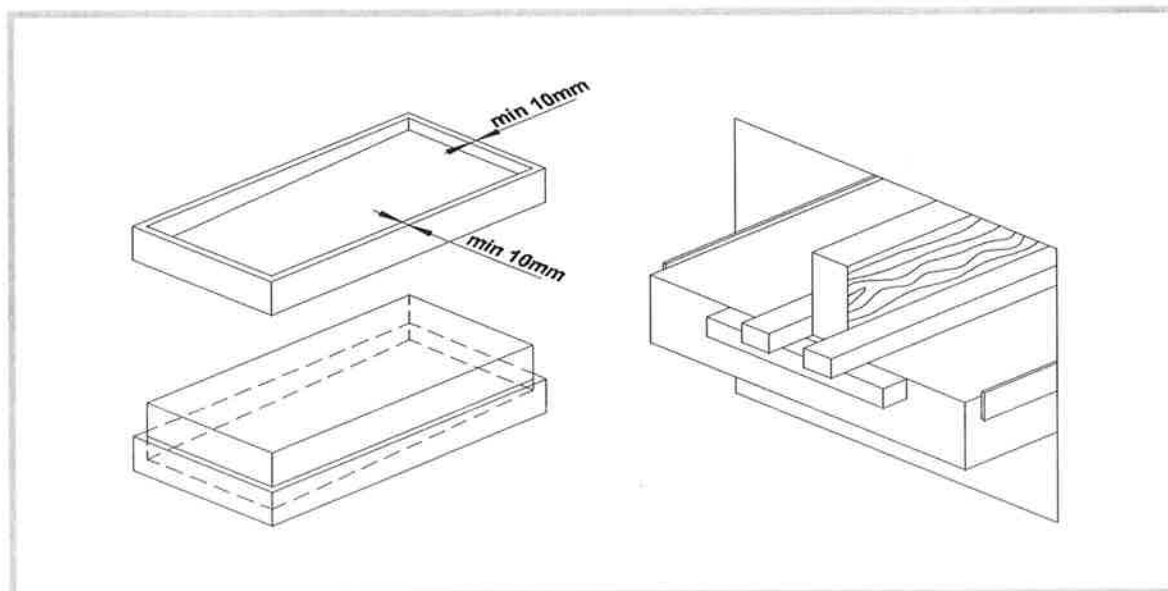


Fig. 17: Examples of tools

4.5 SAFETY OPERATING PROCEDURES

4.5.1 Dimensions of the pieces to be machined

Table 4 shows the min. and max. piece dimensions; in some cases it is possible to machine using suitable machining equipment suitably realized.

	MU	RS53S	RS63S
Max. width	mm	530	630
Min. length	mm	280	280
Max. length	mm	700	700
Min. height	mm	3	3
Max. height	mm	300	300

Tab. 4: Dimensions of the pieces to be machined

4.5.2 Safety measures

- a) Avoid machining woods with too clear defects (breakages, knots, etc...).
- b) Avoid to feed simultaneously different tables with different heights (save for the machines with sectioned roller).
- c) Suit the feeding speed to the piece width and to the removal thickness.
- d) Check if the feeding rollers are free to lift correctly.
- e) Do not stand in front of the machine entry during the machining; do not try and look inside, as chips could be thrown out.
- f) Do not put the hands inside the machine to remove chips or splinters with the machine operating.
- g) If a wood should lock inside the machine, stop the shaft completely, lower the table and remove the wood.
- h) Do not try to remove a quantity of material thicker than the specifications.



5 MAINTENANCE

5.1 MAINTENANCE AND SAFETY

A careful maintenance makes the machine life longer and keeps the machining features unchanged. An important safety factor for the machine and the operator is a general accurate cleaning of the machine and the surrounding area; in the evening, at the end of the works, it is advisable to clean by means of a vacuum-cleaner.

All the installed bearings are life-lubricated, and therefore they do not need any lubrication.

The reducer for the table positioning and the speed gear for the roller advancing are life-lubricated, so they do not need any lubricant change or top-up.

The parts to be lubricated are:

- a) the chain for the roller advancing and the table lifting, to be cleaned with naphtha every 500 hours.



Do not use the machine if the protection and/or safety devices are not efficient.

Before any maintenance intervention, set the main switch to zero and warn about the machine stop by a sign.

5.1.1 Piece anti-ejection device

Each working shift the arms must be examined to check the conditions of the arm contact surface, subject to damages due to shocks and wear, and to check they are freely falling for their weight.

To prevent the possible hardening or locking of some arms, keep them always clean, remove the wooden chips or small pieces stuck between them by powerfully blowing some compressed air.

The resin residues shall be removed washing the whole unit with a brush and some turpentine, then drying by compressed air.

5.1.2 Safety measures

Periodically press the emergency push buttons to check their efficiency.

Check if the motor brake stops the shaft in a max. time of 10 s; if not, to adjust it refer to 5.9.

5.1.3 Tool maintenance

- a) Keep the knives and the gibs clean, to have the best results in the planing.
- b) Resin deposits mixed with wooden dust and small chips must be removed by a hard bristle brush dipped in a suitable and safe thinner.
- c) Do not use any other product, nor synthetic thinners of any kind.
- d) By a jet of compressed air, remove every dirt residue and dry every part.
- e) Clean the working table with a dry cloth.
- f) The knife sharpening is allowed up to a min. height of 20 mm, Table 5; after having reached this value replace the knife.

5.2 TOOLS

5.2.1 Features of the tools that can be installed

STANDARD SPINDLE	MU	RS53S	RS63S
Length (L)	mm	530	630
Max. width (H)	mm	35	35
Min. width (H)	mm	20	20
Thickness (S)	mm	3 ± 0.05	3 ± 0.05
Max. curve allowed on the length	mm	0.1	0.1

Tab. 5: Knife dimension features

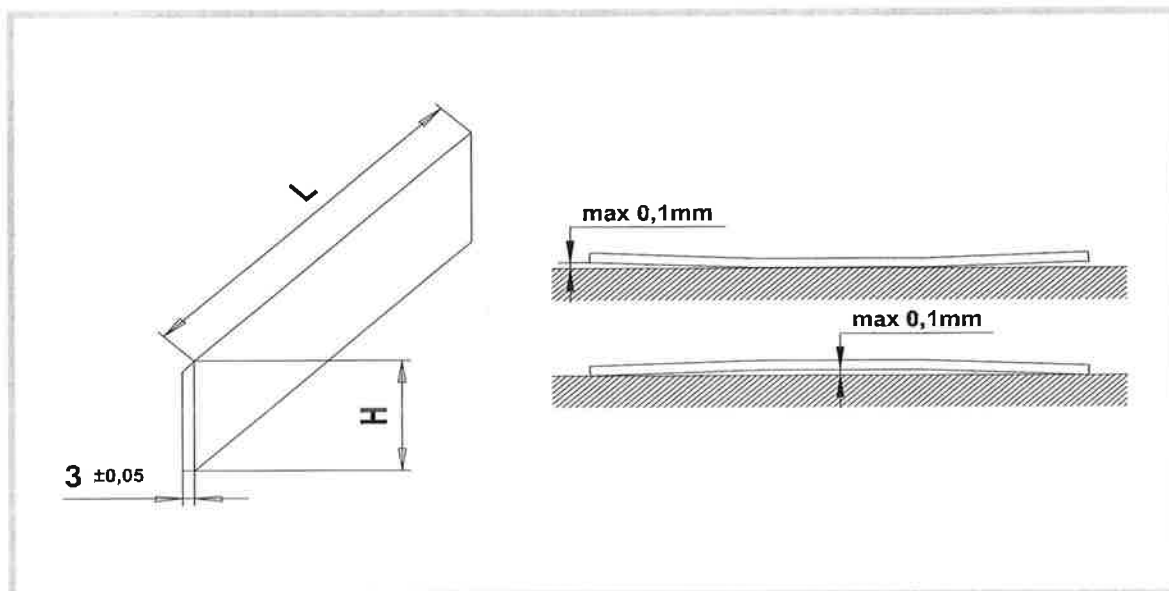


Fig. 18: Standard spindle knives

5.2.2 Standard knife adjustment



Use gloves when handling knives and shafts.

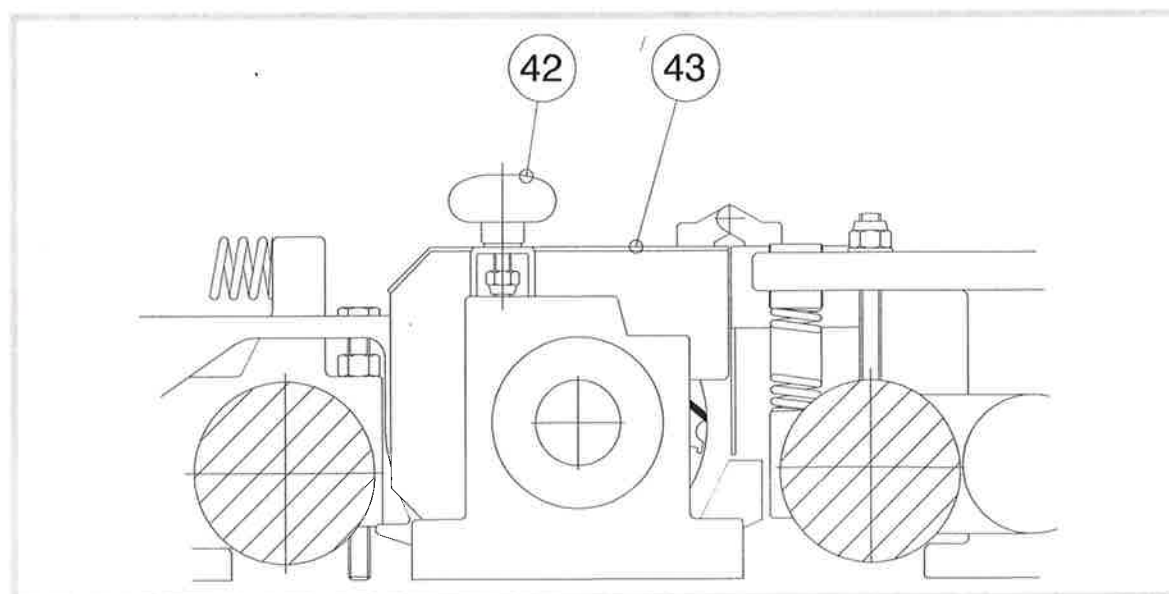


Fig. 19: Plane shaft guard

For adjusting the knives, (Fig. 19) follow the instructions below:

- a) open the upper cover, the interlock microswitch stops the machine operation but enables to use the motor brake release push button;

- b) loosen the knob (42) and open the guard of the plane shaft (43);
- c) rotate the spindle manually in the correct position, holding the brake release push button (7) pressed;
- d) position the knife adjusting tools (39) as indicated and at the ends of the plane shaft;
- e) loosen the screws (37) of the gib (38), in this way the knife (40) will be pushed against the tools (39) by the springs (41) located in the suitable seats on the shaft;
- f) adjust the knife max. projection, that must not exceed 1.5 mm on the radius;
- g) tighten the screws (37) starting from the central one, then tightening on the right and on the left till reaching the ends; use the supplied wrench;
- h) check if all the fastening screws are correctly tightened;
- i) repeat the above steps for the other knives.

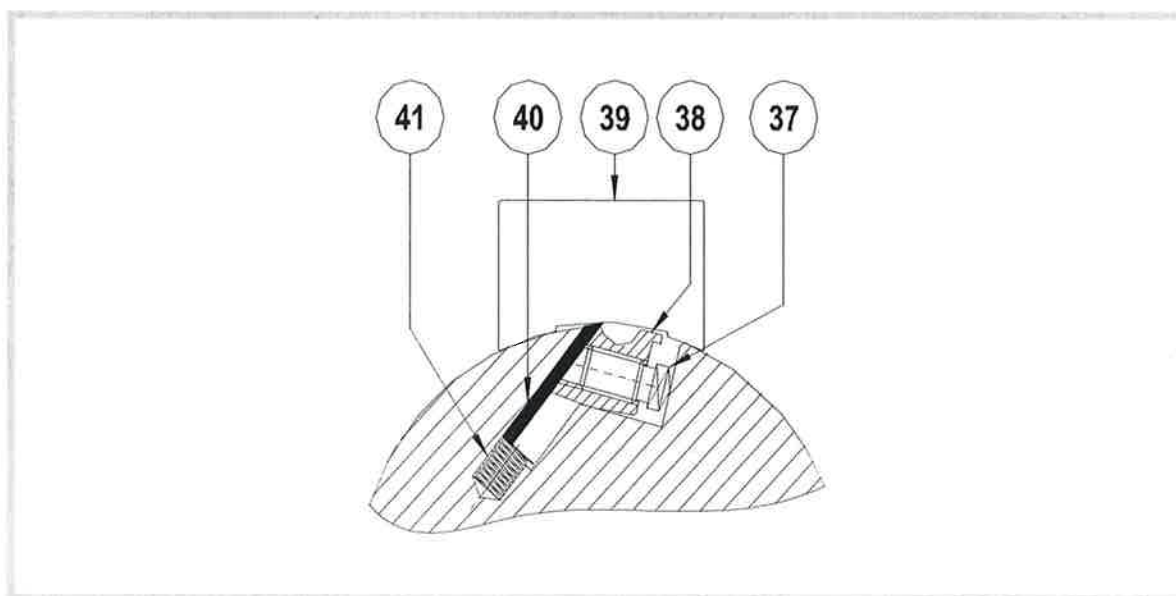


Fig. 20: Standard spindle

5.2.3 Standard knife replacement



Use gloves when handling knives and shafts.

For adjusting the knives follow the instructions below (Fig. 20):

- a) open the upper cover, the interlock microswitch stops the machine operation but enables to use the motor brake release push button;
- b) loosen the knob (42) and open the guard of the plane shaft (43) (Fig. 19);
- c) rotate the spindle manually in the correct position, holding the brake release push button (7) pressed;

- d) clean the gib and its seat;
- e) if it was necessary to remove all the gibs, position them again following the numbers on the gib and on the spindle;
- f) never replace a single knife, but the whole set installed on the machine, to avoid vibration problems due to an incorrect balancing;
- g) replace and/or position again the knife (40) holding it pressed in its seat with a wooden piece (when replacing the knives grease them before the installation); before replacing the knives check if the dimensions and the tolerances are those indicated in Table 5: Knife dimensional features;
- h) tighten the screws (37) to lightly lock the knives, put a oil drop on the screw head to aid the tightening;
- i) position the knife adjusting tools (39) as indicated and at the ends of the plane shaft;
- j) loosen the screws (37) of the gib (38), in this way the knife (40) will be pushed against the tools (39) by the springs (41) located in the suitable seats on the shaft;
- k) adjust the knife max. projection, that must not exceed 1.5 mm on the radius;
- l) tighten the screws (37) starting from the central one, then tightening on the right and on the left till reaching the ends; use the supplied wrench;
- m) check if all the fastening screws are correctly tightened;
- n) repeat the above steps for the other knives.

5.2.4 Replacement and adjustment of the Tersa knives



Use gloves when handling knives and shafts.

For adjusting the knives, follow the instructions below (Fig. 21):

- a) open the upper cover, the interlock microswitch stops the machine operation but enables to use the motor brake release push button;
- b) loosen the knob (42) and open the guard of the plane shaft (43) (Fig. 19);
- c) rotate the spindle manually in the correct position, holding the brake release push button (7) pressed;
- d) release the gib locking the knife with light hammer beats on a wooden wedge;
- e) replace the knife pulling it out;
- f) repeat the above steps for the other knives;
- g) start the knife locking spindle; the spindle start is activated by the centrifugal force.

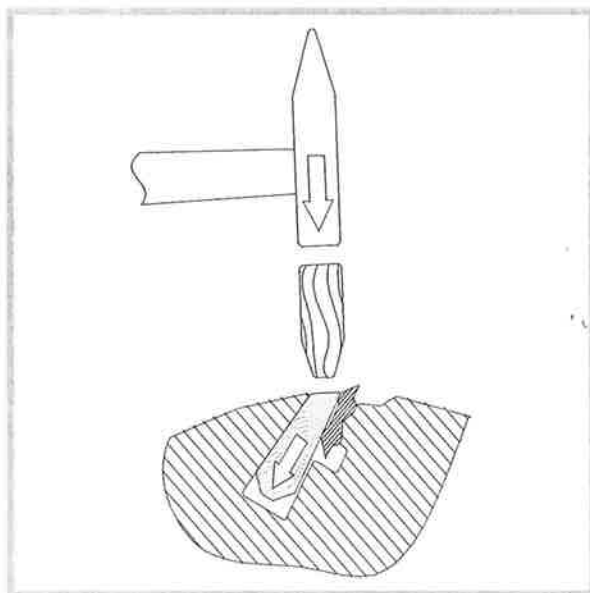


Fig. 21: Tersa and Centrostar spindles

5.2.5 Replacement and adjustment of the Centrostar knives



Use gloves when handling knives and shafts.

For adjusting the knives, follow the instructions below (Fig. 21):

- a) open the upper cover, the interlock microswitch stops the machine operation but enables to use the motor brake release push button;
- b) loosen the knob (42) and open the guard of the plane shaft (43) (Fig. 19);
- c) rotate the spindle manually in the correct position, holding the brake release push button (7) pressed;
- d) release the gib locking the knife with light hammer beats on a wooden wedge;
- e) replace the knife pulling it out;
- f) if necessary, clean the gibs and the spindles: the stuck resin can cause balancing problems;
- g) repeat the above steps for the other knives;
- h) start the knife locking spindle; the spindle start is activated by the centrifugal force.

5.3 TRANSMISSION BELTS

5.3.1 Adjustment of the belt tension

During the initial period of machine use the belts settle on their pulley grooves and the belt tension obviously decreases. After the first 24 ÷ 48 operating hours, adjust the tension, without exceeding not to overload and thus damage the bearings of the motor and of the plane shaft.

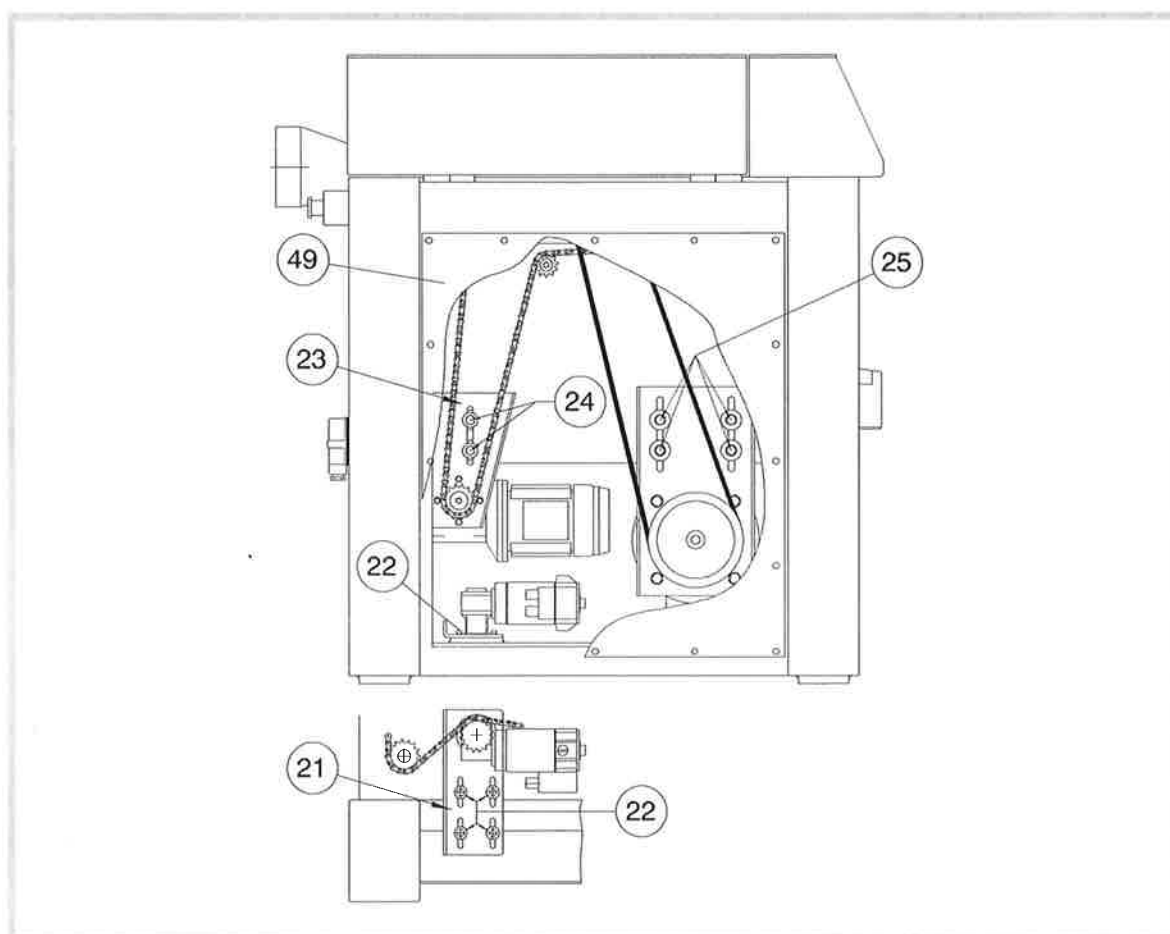


Fig. 22: Belt tensioning

- To reach the belts remove the cover (49) (Fig. 22).
- To tension the spindle driving belts loosen the screws (25) and apply a force to the motor to tension the belts; the tensioning is correct if, applying in the middle of the belt free section a load of 15 ÷ 22 N there is a movement of about 11 mm.
- In case of failure of a single spindle transmission belt replace them all, to avoid the new belt from transmitting during the operation all the power by itself, and thus from being damaged in a very short time.

- d) At the end of the belt tensioning, tighten the screws that had been loosened.
- e) Never match belts of a different make.

5.3.2 Belt replacement

To reach the belts remove the cover (49) (Fig. 22).

Loosen the screws (25).

Using a lever, lift the motor, and keeping it in this position pull the worn belts out and fit the new ones.

Lower the motor and tension the belts as indicated in 5.3.1.

5.4 ADVANCING CHAIN ADJUSTMENT

To reach the belts remove the cover (49) (Fig. 22).

Loosen the screws (24) and correctly tension the chain acting on the ratiomotor support (23).

5.5 ADJUSTMENT OF THE TABLE LIFTING CHAIN

To reach the belts remove the cover (49) (Fig. 22).

After the first 40/60 operating hours check the tension of the table positioning chain.

Loosen the screws (22) to modify the chain tensioning acting on the support (21).

5.6 MAINTENANCE AND REPLACEMENT OF THE FEEDING ROLLERS

To adjust the pressure of the feeding rollers refer to § 4.3.

To adjust the position of the steel or rubber rollers to the working table, refer to § 1.2.4.

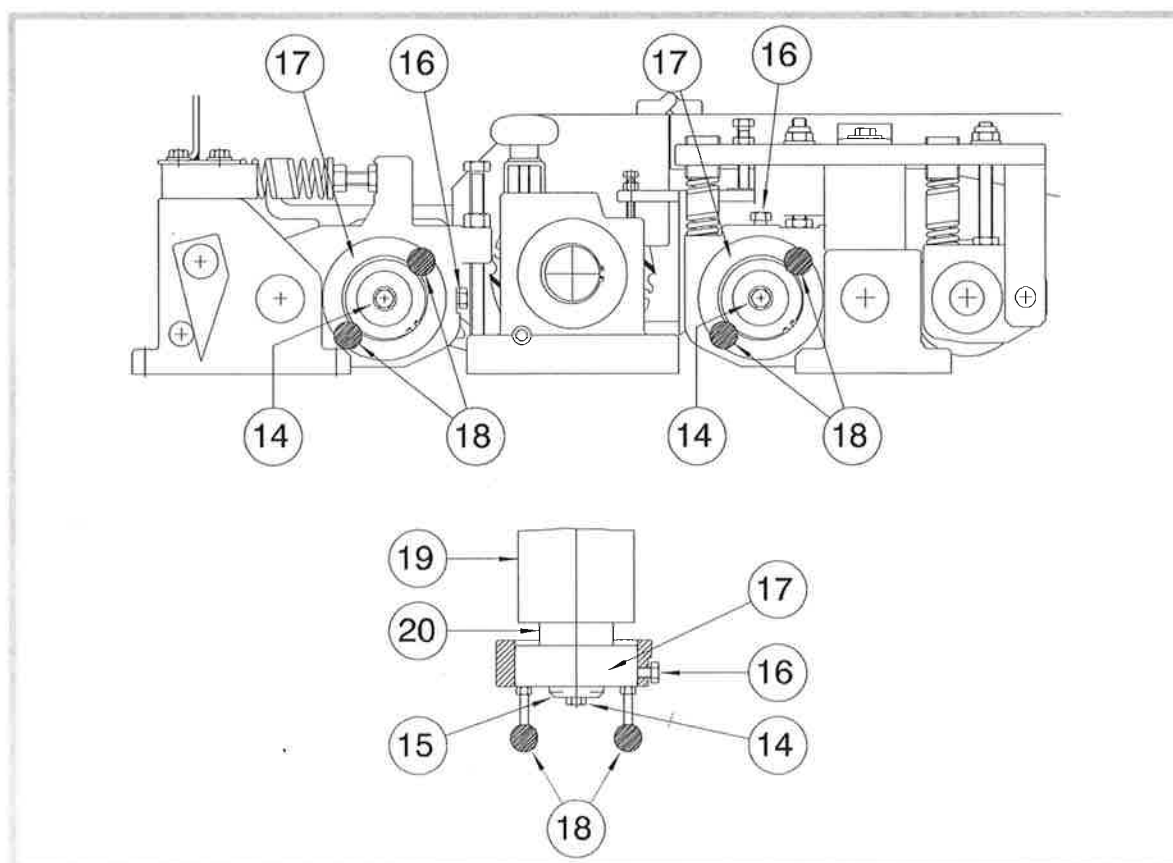


Fig. 23: Replacement of the rubber-coated rollers

Due to the wear, the coating rubber of the feeding rollers gets damaged, and it is necessary to replace them (Fig. 23) following the instructions below:

- lower completely the working table;
- cut the machine off, locking the main switch by a padlock;
- open the interlocked upper cover;
- remove the screw (14) and the washer (15);
- loosen the screw (16) and pull out the support (17) with the bearings, using the knobs (18);
- pull out the spacer (20), then all the roller rubber-coated elements (19) and replace them;
- install again all the components following the reverse order, adjusting the roller pressure as explained in § 4.3.2.

5.7 ADJUSTMENT OF THE REAR PRESSING DEVICE

To adjust the pressure act on the screw (33) releasing the nut (34), the pressure is increased by tightening; after this step lock the screw by tightening the nut.

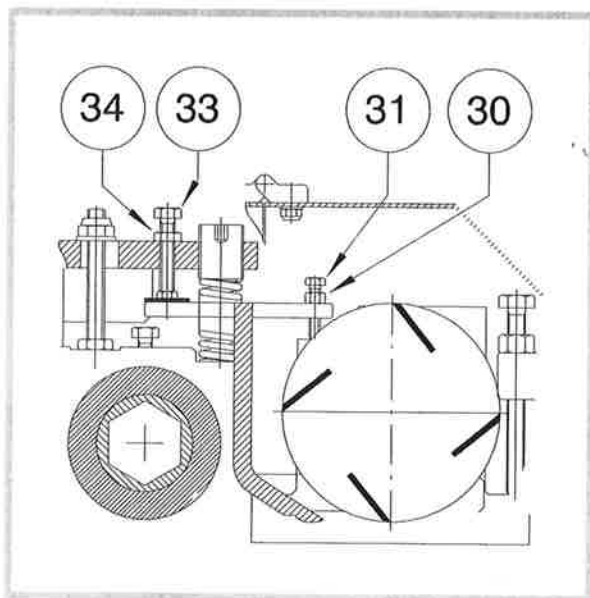


Fig. 24: Adjustment of the rear pressing device

To adjust the height act on the screw (31) releasing the nut (30), by tightening the pressing device is lifted, by loosening it is lowered. In the rest position the rear pressing device must be at the value indicated in the figures 5 and 6 of § 1.2.4.

5.8 ELECTRIC SYSTEM MAINTENANCE

The maintenance of the electric parts must be performed by skilled personnel, with the machine stopped, under authorization of the person in charge with the safety.

Before any intervention, it is indispensable to cut the machine electric supply out rotating the lockable main switch on "0", than disconnect it electrically from the mains.



It's not possible to perform any preventive intervention on the electrical system, but you can only intervene when an element is damaged providing for its replacement; however it is possible to improve the reliability of the electrical parts following some provisions:

- all the motors are supplied with overload cutout, calibrated on the motor rated current; avoid starting and stopping several times consecutively, as this causes a motor overheating and makes the overload cutout ineffective;
- do not remove material requiring a power higher than the installed one;
- replace the faulty components with others having the same features;
- calibrate the overload cutouts at the rated current indicated on the motor plate.

5.9 SELF-BRAKING MOTOR

Every 2 months or 200 stops, periodically check and adjust the braking electro-magnetic device, as the motor braking time must be kept under 10 s.

Detecting longer braking times it is necessary to adjust the entrefer or replace the brake friction material.

5.9.1 Entrefer adjustment

The distance between the electro-magnet case and the movable core (Fig. 25), with the brake not supplied, is called **entrefer** and is adjusted when manufacturing the device and, depending on the size of the installed motor and brake type, it ranges between the values shown in Table 6:

Motor size	MU	Entrefer	Min. thickness of the friction material
TMS 90 – 100 – 112 – 112 – 132	mm	0.2 ÷ 0.5	2
HFV 90	mm	0.25 ÷ 0.5	1
HFV 100 HFV 112	mm	0.3 ÷ 0.55	1
HFV 132	mm	0.35 ÷ 0.6	1

Tab. 6: *Entrefer and friction material*

The entrefer increase depends on the consumption of the friction material (B); this changes the brake performance, such as: noise increase, decrease of the braking capacity with following longer braking time, problems for the brake electric release.

When the entrefer reaches a value higher than the max. allowed one, it is absolutely necessary to take this value back to the min. allowed one according to the type of motor installed on the machine.

For adjusting the entrefer, remove the air conveyor (18), then act on the locknut (16) till reaching the suggested entrefer, measuring it by a feeler gauge.

While adjusting, check the wear condition of the friction material; if the wear is too much, it is necessary to replace the movable anchor – brake disk (12).

The correct brake operation cannot be ensured when operating in oily environments or under extreme temperatures, if the proper guards are not used.

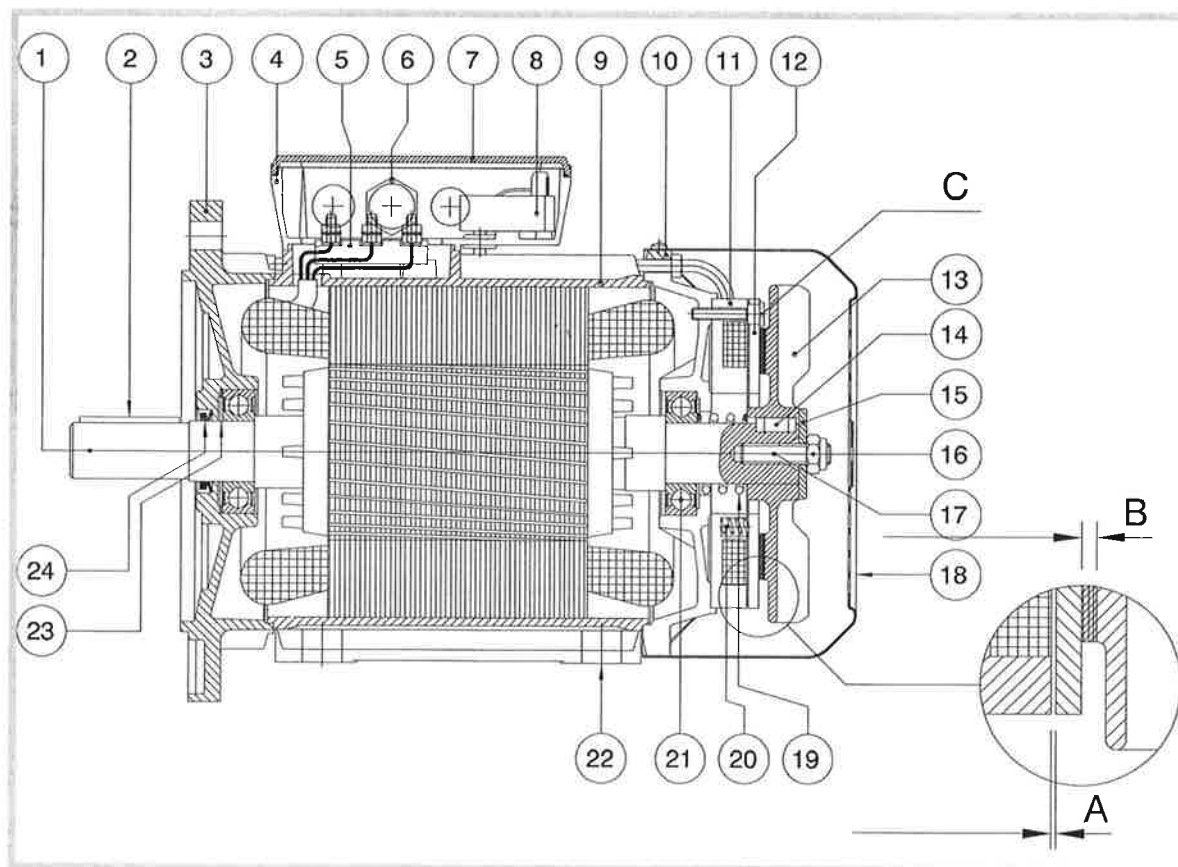


Fig. 25: Light self-braking motor

5.9.2 Replacement of the movable anchor - brake disk (12)

- Remove the air conveyor (18);
- loosen the locknut (16) for adjusting the braking torque;
- remove the washer (15);
- pull the cooling and braking fan (13) out;
- loosen the three screws (C) fastening the electro-magnet;
- remove the movable anchor-brake disk (12) and replace it;
- install again all the components following the reverse order, adjusting the entrefer value as explained in § 5.9.1.

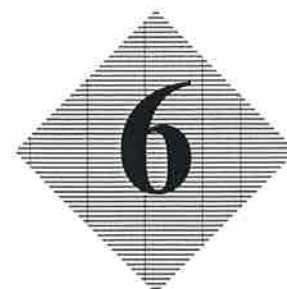
5.9.3 Replacing the electro-magnet (11)

- Open the terminal board cover (7) and disconnect the electro-magnet (11) from the feeder-rectifier (8);
- remove the air conveyor (18);
- loosen the locknut (16) for adjusting the braking torque;
- remove the washer (15);

- pull the cooling and braking fan (13) out;
- loosen the three screws (C) fastening the electro-magnet;
- remove the electro-magnet (11) and the movable anchor-brake disk (12) and replace them;
- install again all the components following the reverse order, adjusting the entrefer value as explained in § 5.9.1.

RS53S - RS53S





6 WHAT YOU MUST DO IF...

6.1 FOREWORD

The machine transport, unloading and positioning, an incorrect use or a poor maintenance may result in problems that can be solved according to the following points.

Symbol key:

- Interventions that the user can do.
- Interventions that must be made by the skilled technical staff of the (inner or outer) service.

6.1.1 The spindle overload cutout does not reset

CAUSE	REMEDY	
Check the power supply.	Check that the power supply mains to which the machine is connected supplies current.	■
One of the emergency push buttons is pressed	Rotate the emergency push button to reset it.	■
Overload.	Wait 10–30 seconds before resetting it.	■
One of the protection switches of the transformer has intervened.	Reset it.	■
Overload cutout faulty.	Replace it.	■

6.1.2 The spindle motor rotates always braked

CAUSE	REMEDY	
The brake coil is faulty.	Replace it.	●
The brake rectifier is faulty.	Replace it.	●

6.1.3 The overload cutout of the spindle motor has intervened

CAUSE	REMEDY	
Overload due to too much removal.	Reduce the removal depth.	■
Overload due to an excessive feeding speed.	Reduce the feeding speed.	■

6.1.4 Piece surface marking

CAUSE	REMEDY	
Insufficient suction	Check the connection with the suction system	■
Chip discharge container in the spindle clogged	Clean the chip discharge in the spindle	■

6.1.5 Ledging close to the piece exit edge

CAUSE	REMEDY	
Excessive pressure of the feeding roller at the entry	Adjust the load pressure	■
First roller on the working table too projecting	Decrease the roller projection	■

6.1.6 Planed surface not parallel to the supporting plane

CAUSE	REMEDY	
Knives not aligned to the plane	Adjust the knives	■

6.1.7 Ledging close to the piece entry edge

CAUSE	REMEDY	
Excessive pressure of the feeding roller at the exit	Adjust the load pressure	■
Second roller on the working table too projecting	Decrease the roller projection	■

6.1.8 Tears on the piece surface

CAUSE	REMEDY	
Too high feeding speed	Reduce the feeding speed.	■

6.1.9 The piece advances with irregular speed

CAUSE	REMEDY	
Insufficient pressure of the feeding rollers	Adjust the load pressure	■

Dirty working table	Clean the table with naphtha	■
Too high feeding speed	Reduce the feeding speed	■
Excessive removal	Reduce the removal	■

6.1.10 Electronic value display calibration



The calibration must be made by skilled staff, as the steps must be made with the machine live to enable the movements through the positioning ratiomotor; see § 4.3.1.

The following steps involve the calibration of the electronic display:

- display replacement;
- reading system replacement;
- interventions on the working table or replacement.

For a correct calibration of the display operate as follows:

- connect the display to the machine supply and to the measuring system as indicated in the wiring diagram (if, moving the table downwards, the display increases the value instead of reducing it, revert the wires on the terminals 1 and 3 of the display);
- position the working table so as to flat a board 10-mm high, and measure the obtained thickness with a gauge;
- remove with a screw-driver the transparent plastic panel covering the instrument (Fig. 26);

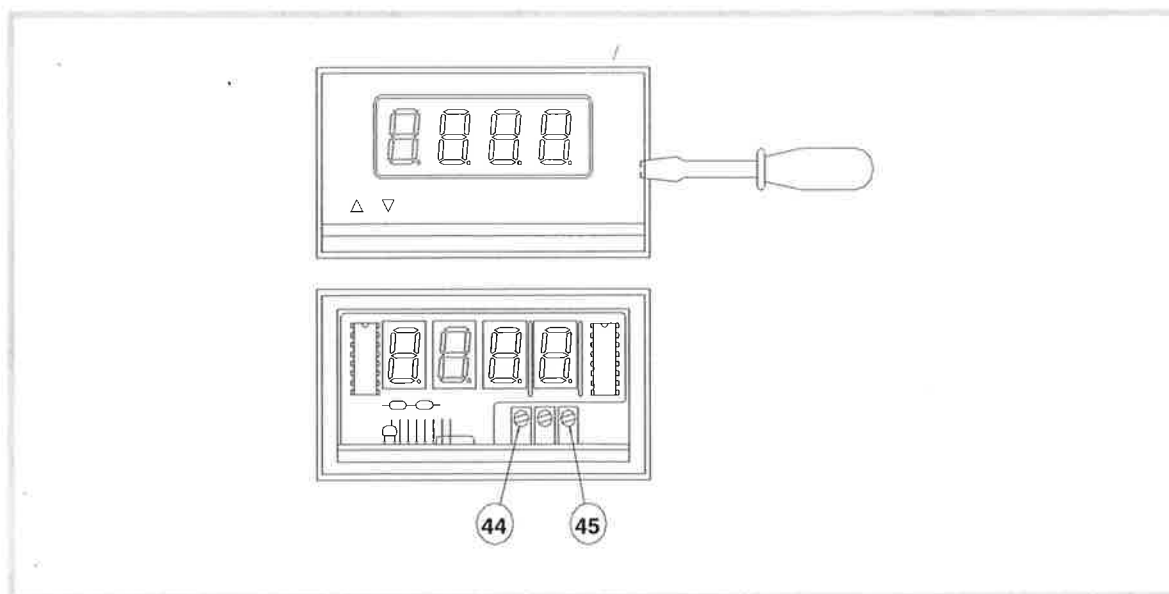


Fig. 26: Electronic display

- operate on the trimmer (44) making the displayed value correspond to the measured one;
- lower the working table so as to flat a thicker board and measure with a gauge the obtained thickness;

- operate on the trimmer (45) making the displayed value correspond to the measured one;
- then check the min. and the max. working value.

The calibration ends when passing from the min. value to the max. one and vice versa it is not necessary anymore to correct the displayed value.

After the operation, close again the display transparent cover.

6.1.11 Replacement of the table positioning ratiomotor

This intervention must be made by skilled staff, as the relevant steps must be made with the machine side protection cover (49) removed to enable the maintenance.

Set the machine main switch on "0" and place a plate on the machine to indicate the machine maintenance condition.

Remove the side protection cover (49).

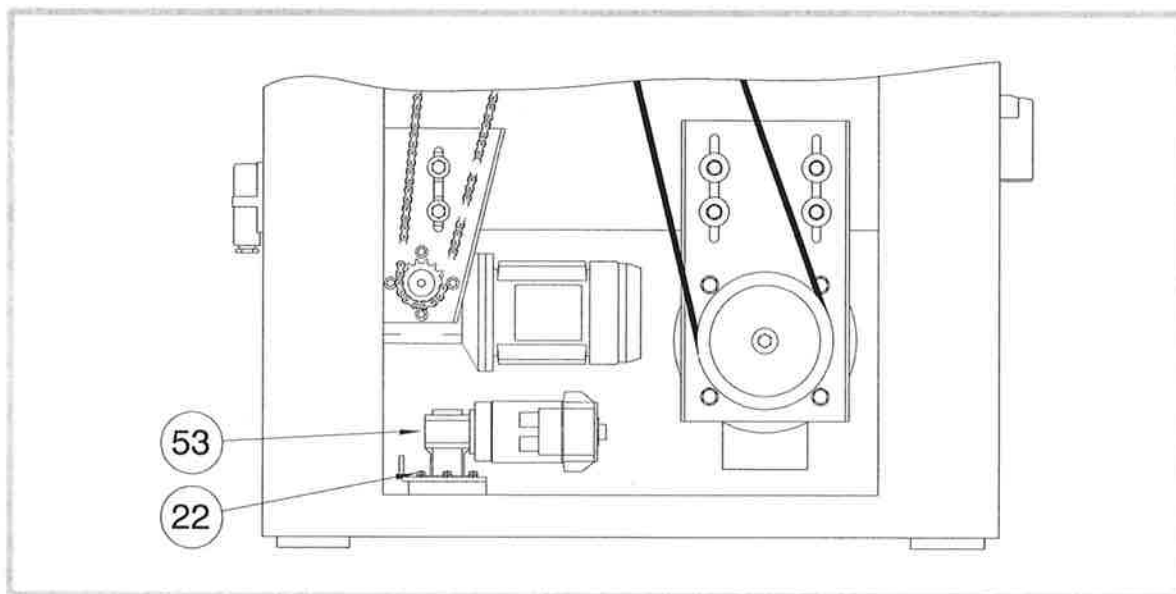


Fig. 27: Table positioning ratiomotor

The motor and the ratiomotor (53) form a single piece, and can be disassembled only by the manufacturer of the product itself.

Disconnect the motor cable from the terminal board and loosen the screws (22), then remove the ratiomotor levering with a screw-driver under the connection flange, then replacing it with a new one and repeating the steps in the reverse order.

At the end, close again the cover (49).

6.1.12 Replacement of the feeding ratiomotor



This intervention must be made by skilled staff, as the relevant steps must be made with the machine side protection cover (49) removed to enable the maintenance.

Set the machine main switch on "0" and place a plate on the machine to indicate the machine maintenance condition.

Remove the side protection cover (49).

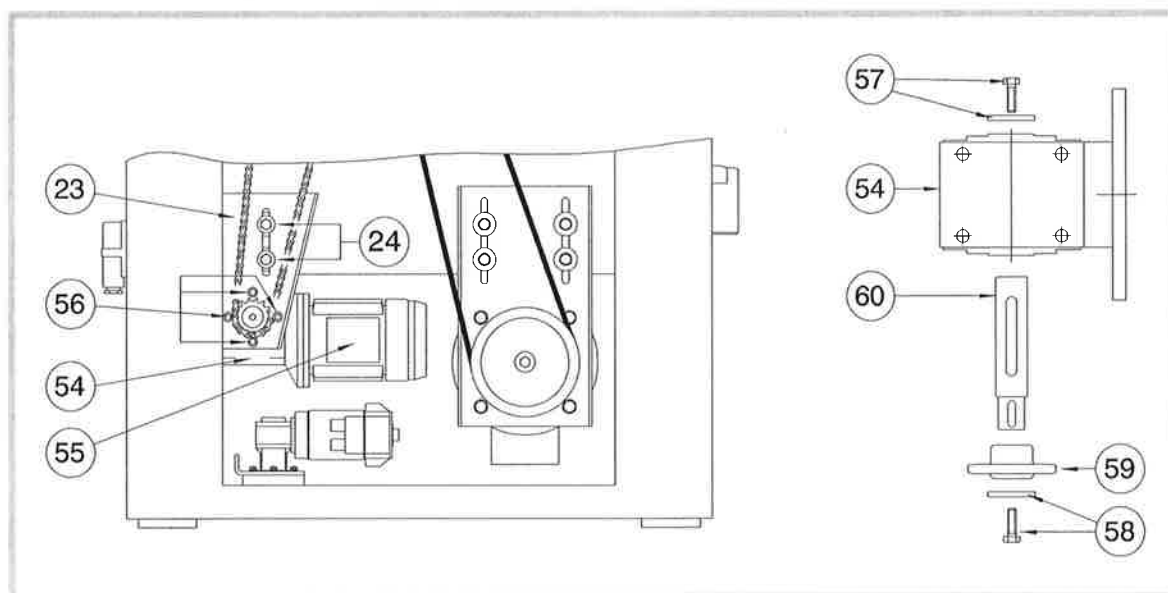


Fig. 28: Feeding ratiomotor

Loosen the screws (24) so as to clear the support (23) and remove it with the motor (55) ratiomotor (54); then lay the whole on the ground.

Remove the motor (55) from the ratiomotor (54) extracting the connection screws and levering with a screw-driver under the connection flange.

Remove the ratiomotor from the support (23) loosening the screws (56), then remove the screw and the washer (57), and pull out the shaft (60) on which the pinion (59) is installed, seated by the screw and the washer (58).

Replace the ratiomotor with a new one, and repeat the steps in the reverse order.

At the end, close again the cover (49).

RS53S - RS53S





7 TECHNICAL AND ENVIRONMENTAL DATA

7.1 TECHNICAL DATA

	<i>MU</i>	<i>RS53S</i>	<i>RS63S</i>
STANDARD FEATURES			
STANDARD SAC spindle with 4 knives		☐	☐
Plane shaft starting with automatic switch		☐	☐
Automatic motor brake		☐	☐
1° Steel feeding toothed roller		☐	☐
2° Smooth steel feeding roller		☐	☐
3° Smooth steel feeding roller		☐	☐
Feeding rollers on bearings		☐	☐
Electronically variable feeding speed		☐	☐
Interlocked feeding drive		☐	☐
Table lifting with 4 accuracy screws protected against the dust		☐	☐
Positioning of the motor-driven table		☐	☐
Table end positioning		☐	☐
Table value electronic display		☐	☐
Screw life lubrication		☐	☐
Table guide on self-lubricating bushes		☐	☐
Sectioned entry pressers		☐	☐
Suction hood		☐	☐
Entry and exit protection bands		☐	☐
TECHNICAL DATA			
Table dimensions	mm	1050 x 530	1050 x 630
Useful working width	mm	530	630
Min. working height	mm	3	3

	MU	RS53S	RS63S
Max. working height	mm	300	300
Max. passage depth	mm	8	8
Piece min. length	mm	300	300
Knife-holding spindle diameter	mm	120	120
Shaft rotation speed	min ⁻¹	4760	4760
Knife number	No.	4	4
Knife dimensions	mm	530 x 35 x 3	630 x 35 x 3
Feeding speed	m/min	4 ÷ 16	4 ÷ 16
Suction inlet Diameter	mm	160	160
PACKING – TRANSPORT			
Overall dimensions for the transport:			
Width	mm	1065	1165
Length	mm	1135	1135
Height	mm	1230	1230
Case packing dimensions:			
Width	mm	1110	1210
Length	mm	1220	1220
Height	mm	1390	1390
Weight	kg	930	1030
Weight with package	kg		
ELECTRICAL DATA			
Spindle motor power	kW	4	5.5
Feeding motor power	kW	0.55	0.55
Table positioning motor power	W	150	150
Three-phase voltage	V	230/400	230/400
Connection voltage	V	400	400
Frequency	Hz	50	50
Current	A		
Low voltage circuit with transformer	V	230	230
SUCTION DATA			
Suction air min. speed:			
for dry chips	m/s	20	20
for wet chips (humidity ≥ 18%)	m/s	28	28
Capacity	m ³ /h	1,800	1,800
Vacuum	mmH ₂ O	720	720
ENVIRONMENTAL DATA			
Storage temperature	°C	0° ÷ 50°	0° ÷ 50°
Operating temperature	°C	5° ÷ 40°	5° ÷ 40°

	MU	RS53S	RS63S
Max. relative humidity		90%	90%
Max. elevation	m	1,000	1,000
STANDARD ACCESSORIES			
Use and maintenance handbook	☐	☐	☐
Wiring diagram	☐	☐	☐
Spare part catalogue	☐	☐	☐
Set of wrenches for the maintenance	☐	☐	☐
Knife installation/adjustment device	☐	☐	☐
ACCESSORIES upon REQUEST			
MONOBLOC spindle with 4 TERSA knives		■	■
TERSA spindle		■	■
CENTROSTAR spindle		■	■
Spindle motor power	kW	5.5 - 7.5 - 10	7.5 - 10
1° Steel feeding toothed roller, sectioned		■	■
2° Rubber-coated smooth feeding roller		■	■
3° Rubber-coated smooth feeding roller		■	■
Pneumatic device for the pressure adjustment on the feeding rollers		■	■
Machining table with 1 adjustable roller		■	■
Machining table with 2 adjustable rollers		■	■
Table positioning electronic programmer		■	■
One set of knives (4) Standard		■	■
One set of knives (4) TERSA		■	■

Key: ☐ = Standard ■ = Optional

7.1.1 OVERALL DIMENSIONS

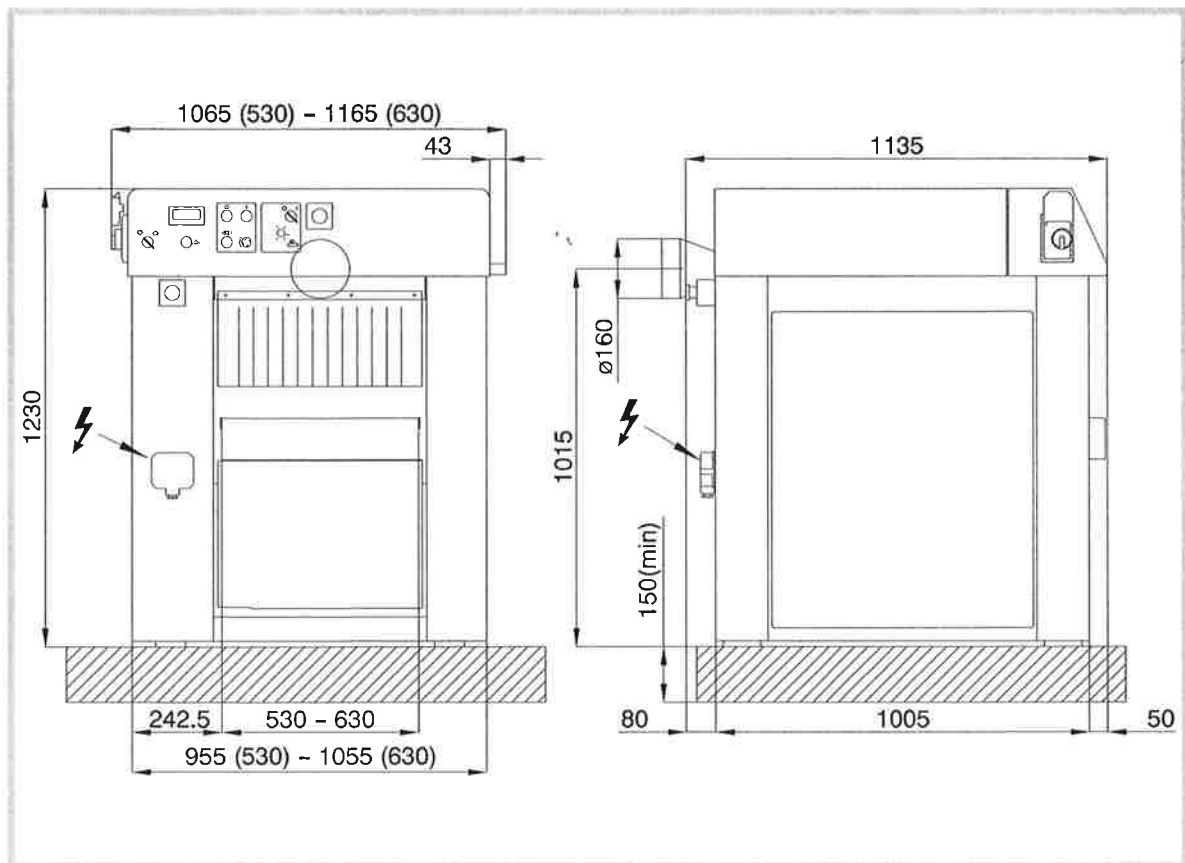


Fig. 29: Overall dimensions

